Totality as a Category

An Epistemological Investigation

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Chapter I

The Position of the Doctrine of Categories in Philosophy

1. Categories are fundamental concepts which are contained in the presuppositions on which all scientific research rests, and which can be exhibited by analysis of those presuppositions. But for a full elucidation of their nature it is not enough to start out from the science existing at a given time. It must also be shown how they have worked their way forward through the history of science, — how they arise as forms of thought-work, — and how they can disappear again when they are no longer expressions of labouring thought. And from such a historical consideration one must go back again to a psychological investigation of the development of thinking in ordinary human conscious life, of which scientific thinking is a peculiar formation conditioned by determinate tasks. The fundamental concepts with which science works must be psychologically possible, and their first origin must be findable in involuntary thought-life. The difference between the psychological and the epistemological consideration of the categories consists in this, that the former finds possibility for unscientific as well as for scientific thinking, whereas the latter shows that under certain given conditions there is only one single narrow road to understanding. Of our thoughts it holds that many are called, few chosen, — indeed, we must add as psychologists, that many are uncalled.

In epistemological works one will find different weight laid on fundamental concepts and on fundamental propositions. Some epistemologists keep solely to the fundamental propositions, the propositions on which scientific grounding rests. And since propositions that are presuppositions for all grounding naturally cannot themselves be grounded, the fundamental propositions are often presented as resting on a choice, a leap, an act which in and for itself might just as well have fallen out in quite another direction. The first principles of science are, on this conception, not themselves an object for science. Yet it is for the most part conceded by those who hold this conception that the choice is bound within a certain circle of possibilities, and that this circle of possibilities is given with the nature of human thinking, such as this once and for all is constituted. It is psychology's task to investigate this factual constitution, and the question then presents itself how the different standpoints from which epistemology and psychology assess human cognition stand to one another. But even before we take up this question it will be clear that there is such a reciprocal relation between fundamental concepts and fundamental propositions, as in general between concepts and judgments, that the investigation of the single concepts which the fundamental propositions contain will throw light on the fundamental propositions themselves. Advancing clarity becomes possible at all only by our investigating and determining, with express consciousness, the concepts contained in our judgments, just as on the other side we investigate and determine the kind of connection that is asserted in the judgments between the concepts. The two investigations supplement one another. Advancing clarity in the determination of concepts is the condition for advancing clarity in the understanding of the judgments into which the concepts enter, and conversely. Whether one lays the greatest weight on the one or the other investigation will depend on the special task one sets oneself. But for a complete reckoning neither of the two standpoints can be dispensed with. To this it comes in addition that the type of thought is the same in concepts as in judgments. The judgment is a connection of concepts, but the concept is a connection of marks. It is the same fundamental property of our thinking that expresses itself in both places, in the concept more as structure, in the judgment more as function.

Whereas by keeping merely to science's first presuppositions one arrives at an uncompromising theory of postulates, which places all scientific thought-work in a distant relation, or even in sharp opposition, to all other spiritual work, one will, by investigating the fundamental concepts as well, be able to be led to an understanding of the relation of scientific thought-work to the life of the spirit as a whole. And since it is precisely one of philosophy's tasks to investigate this relation, it will be of importance to attempt a determination of the position of the doctrine of categories within philosophy.

2. When does science begin? And how does scientific consciousness stand to the thought-life that unfolds before and alongside the scientific, in greater or less independence of it?

For the elucidation of these questions it must first be held fast that a common type makes itself felt in all human thought-life, the prescientific, the scientific and the unscientific alike.

Practical life itself, its necessities and its tasks, leads to thinking. In order that a man may attain the ends he strives after, there are certain means that must be procured. Here man learns in fact and in practice what necessity is, that is, he learns to know the inexorable relation between ground and consequent. So long as he holds fast to the end, so long must he also concede the necessity of means, and finally of certain determinate means; and he learns conversely to see that the attainment of the end is a consequence of the application of certain means. He learns besides that when the same necessities and the same outer conditions repeat themselves, the same means must be resorted to. A Central Australian, for example, gathers plants, threshes them with a stick, sifts the seed, roasts it, crushes it on stones and eats it: a series of actions which, every time he is hungry, must be performed in the same way and in the same order. At bottom, science's first presuppositions may be said to be present already in this series-formation.

In the most recent time particular weight has been laid on the influence of social life on the first development of thinking. In the tribe's division into different groups or classes there lay a model for all classification, and thereby for all logic. We still speak, after all, of genera and families in the division of natural objects, in spite of the fact that eager theorists of heredity find false analogies here. The representation of time and the division of time received their development through the rhythmical course and regular recurrence of religious feasts and ceremonies. Our almanac still indicates the scheme of the church year. And religious worship demanded that liturgical acts be performed in a quite determinate manner, the same in every single case, if the result was to be attained. Here, then, was inculcated the proposition that like must be produced by like. The division of space received its clarity by reason of the religious significance of the different directions and the different places.[1]

[1]Émile Durkheim has, especially in his work *Les formes élémentaires de la vie religieuse* (1912), made

That the social conditions under which men develop exercise a very great influence on their way of thinking and their forms of thought is beyond doubt, even if not everything can be explained along this road. Social influence is constantly broken against the practical experience individuals can make on their own account, face to face with those conditions of life which also gain influence on the conduct of society as a whole.

Whether one will now lay the greatest weight on individuals' practical experiences or on the influence of social institutions, it is a fact that at each given time and among each single people there appears a certain conception of the world, involuntarily and in part unconsciously developed, which is built after the same main forms as those that appear in a more stringent and special manner in scientific thinking. It bears witness to a need and a capacity to order and interpret observations and traditions and thereby to bring them into a certain connection. The thought-work that is done here proceeds without clear consciousness. No attention is directed to the single transitions of thought or to the warrant for connecting, in a certain determinate way, the single elements that observation and tradition offer. To use the terminology I have employed elsewhere: it is not analytic and synthetic intuition, but practical and concrete intuition, that is present here.[2] So-called sound common sense operates throughout with these last-named kinds of intuition. Through them it can, to use MEYERSON's expression,[3] give "a first, very clumsy sketch of a scientific and metaphysical system". — In language, in the words language forms and the word-combinations it shapes, such a system is already contained, one that constantly makes its influence felt on the scientific forms of thought.

The origin and development of science, as the history of science shows it to us, really makes the question of its first beginning fall away entirely, since it turns out that its first germ is present as soon as men begin to think. At all stages and under all conditions a certain opposition makes itself felt between something given, something presented, and its interpretation, between the single subject-matter and the connection into which it is fitted. Progress consists in there coming to be an ever closer and stricter relation between observation and interpretation. More and more one becomes conscious that all

this way of looking at the matter prevail and sharpened it into the assertion that all categories are of religious-social origin. Cf. my treatise *Sociologi og Religionsfilosofi* (Vor Tid, 1914–15), (in French in *Revue de la Métaphysique et de Morale*, Nov. 1914).

[2]See *Om Begrebet Intuition* (Vid. Selsk. Oversigt 1914), p. 77.

[3]MEYERSON has discussed the concept of *sens commun* in an excellent manner in his book *Identité et Réalité* (1908), Chap. XI.

interpretation must in the end itself be fetched from observations, and that the fitting of the single subject-matter as a member into a whole connection must take place after exact testing both of the single observation itself and of the other observations with which it is connected. As will be shown in what follows, it is here especially the difference between identity and analogy, and the limit of analogy's warrant, that one must have an eye for.

3. It is psychology's task to find the common type for all thought-life, the scientific as well as the prescientific and the unscientific.

This type can in its generality be designated as a whole-formation. Even the simplest sensations of sense show themselves to stand in connection with one another, so that the single one cannot be understood, either as regards degree of strength or as regards quality, except by being seen as a member in a whole series or group of sensations. The temporal relation between the different sensations is already decisive for the degree of independence with which they appear. To this it comes in addition that reproductions of earlier sensations unite themselves more or less closely with the newly emerging sensations, often from the very hour of their birth. Thereby recognition, partial perception and sense-illusion become possible. Where the reproduced elements gain the decided upper hand, memory and imagination appear as a whole-formation of a higher order. Through association, displacement and comparison, intuitions of sense, of memory and of imagination can undergo alterations, articulations, without the character of intuitability disappearing. Cooperating in all these whole-formations, from first to last, are practical motives, need and impulse, pleasure and displeasure.

Characteristic of the whole-formations here spoken of (which lead to what I have called concrete and practical intuition) is the involuntary manner in which they come to be. From this follows the self-evidence and originality they seem to possess. The elements of which they are built up are not themselves accessible to us in the same way as the elements out of which we form concepts, judgments and inferences with full consciousness, since the marks of the concept, the members of the judgment and the premises of the inference are then given in advance of the new whole-formations into which they enter. Thereby what I have called analytic and synthetic intuition stands in opposition to concrete and practical intuition. Now it is in fact not all concepts, judgments and inferences that we form with full consciousness; they can also arise by involuntary whole-formation. But if concepts, judgments and inferences are to have scientific significance, they must be capable of being formed anew, in such a way that full and clear consciousness is present at each single transition from thought to thought (analytic intuition) as well as at each single connection of thought with thought (synthetic intuition). Yet even in the involuntary formation of concepts, judgments and inferences, reflection appears as something different from immediate intuiting (in sensation, memory and imagination), inasmuch as a more or less distinct transition and comparison takes place between the elements that are brought into connection, whereas the elements out of which immediate intuitions are formed can be traced only indirectly and in part by the road of analogy.

Since the formation of concepts takes place by judging, and since the inference is the formation of a judgment out of other judgments, we can in this connection keep to the judgments.

The judgment is formed in the first instance on the basis of immediate intuiting. It states a determinate relation between elements in the content of the intuition, elements which in the intuition itself stand in continuous connection or are perhaps even fused with one another. The judgment "the table is square" is formed on the basis of an intuiting in which the property "square" was already contained ("lay"), but without being specially brought forward and without being placed in relation to the table as a whole, that is, as the sum-total of all its properties.

But why are judgments formed at all, — why do we not remain at intuiting? It is surely not, as some authors seem to conceive the matter, a kind of fall into sin that has taken place here. It is the richness and manifoldness of observation, and memory's attempt to span this richness, that make another kind of whole-formation than immediate intuiting necessary. The primary form of the whole is burst open, and a new one is sought.

It can be difficult to draw a sharp boundary between intuition and judgment. As already indicated, the intuition can be altered, articulated. This happens by single parts pushing themselves into the foreground while others recede. In the case of the table, for example, the colour can make itself specially felt in preference to the figure, and in the figure itself the contour in preference to the form of the surface.[4] Such displacements go on unceasingly in our sensations, memories and imaginings. But a judgment is formed only when we become conscious of these transitions instead of being wholly absorbed

[4]Cf. Edgar Rubin: *Synsoplevede Figurer.*

in the intuitions that are present in the single moments and that succeed one another. Now as regards this consciousness there may indeed be many transitions and degrees; but the end-points nevertheless stand in characteristic opposition to one another, and it would be unwarranted to transfer to intuition everything that holds of judgment. There is only analogy between them, an analogy grounded in their being two different ways in which psychical energy, that is, the need and the capacity for whole-formation, can express itself.

The founders of the doctrine of categories, Aristotle and Kant, rightly saw that the scientific fundamental forms of cognition must be findable by analysis of the judgments that are passed. What they have not seen clearly enough is that the psychical energy at work in the formation of judgment also works, under other conditions, in other whole-formations. Only through the continuity and analogy that make themselves felt between the primary whole-formation and the judgment is it understood how scientific consciousness stands in close connection with the life of consciousness as a whole and is a more special formation of it. But it is also precisely only a relation of analogy that is present, as between two examples of the same concept. Since judgments can be derived from, built upon, intuitions, one may readily say that they are "contained" in them or "lie" in them. But that is figurative speech. For because a judgment is logically possible, it is not thereby said that it is psychologically possible; and in any case it is a conversion into another form that takes place in the transition from intuition to judgment, a transition that presupposes certain determinate conditions. One must not confuse the immediate and involuntary security with which the mind rests in intuitions of sense and of memory with the assurance that is first the work of reflection. Such a confusion is committed by Heinrich Maier[5] when he declares: "Elementary judgments lie in every observation, every memory-image, every cognitive representation of imagination, in so far as all these representations have real objects, real events, states, things and so forth for their object". And when he adds: "The alleged images of observation, memory and imagination, which are supposed to be indifferent to the opposition between true and false, valid and invalid, are fictions that never and nowhere are real", — then I can only agree with him in so far as that immediate security cannot be called indifference. Strictly speaking one can speak of

[5] *Psychologie des emotionalen Denkens* (1908), p. 2. — Cf. on this whole question *Den menneskelige Tanke*, p. 53–59; 74 f.; 99–103.

indifference only where both members of the opposition are known and stand as equally valid. But this acquaintance is not present from the first. There has not yet been any eating of the tree of knowledge. No doubt has yet stirred; the state is positive through and through. When the immediate security is broken through contradiction, or through the demands for closer determination that new observations make, then reflection begins its work of producing new whole-formations, and then the judgment can replace the intuition. Only then does application arise for the concepts true and false as opposites.

4. The difference between elementary, involuntary thought-work and scientific thought-work can be elucidated from three different sides.

a. Reflection will first of all seek to order the different subject-matters it distinguishes according to their mutual relation. Relation, determinateness by relation, is therefore (as the history of the doctrine of categories also shows) the most conspicuous category. But orderings determined by relation can be of different significance for the development of cognition. Through a series of stages reflection advances from purely external orderings, which cannot lead further, to orderings that make it possible to derive new relations from those already present. There is a whole series of stages from chaos to rationality, and rationality can itself have different degrees. These stages are elucidated by the series that can be formed by the ordering of the subject-matters. A survey of these series I have given in "Den menneskelige Tanke" (p. 162–207); but I come back to them here in all brevity and from a somewhat different point of view.[6]

α. If we try to think for ourselves the conditions most unfavourable for human thinking, we stand before the logically irrational. We stand before such a relation between given subject-matters that they are all equally different from one another, and that they can therefore be placed in any order whatever. One may push out any member whatever without the relation between the remaining members being altered, and without anything following from it at all. Such a series I have called a chaotic difference-series. It does not appear in immediate observation. For already, purely psychologically, the order of the addends is not a matter of indifference. Our mental states are determined, both as regards degree and as regards quality, by their place in the series of our lived experiences. But there can be all possible approximations to such a logical irrationality, which differs from

[6] As an appendix to the present treatise there is printed a list of the categories, essentially as I have presented them in the book named. And there are also set out the different series-formations that express the transition from prescientific to scientific consciousness.

mathematical irrationality in this, that it cannot, as the latter can by its very manner of coming to be, lead to a reduction of the differences. We know that $\sqrt{2}$ lies in the series of numbers between 1 and 2 and nowhere else, and within the interval named the place can be determined ever more exactly the more decimals we calculate. The logically irrational, by contrast, means a halt.

The next step is that series are formed whose members lie fast, while the relations of difference vary from member to member. Such an indeterminately varying difference-series does not permit elimination without the members between which the elimination takes place coming into another relation to one another than before. But the advance consists in this, that clear survey has replaced the chaotic unrest.

A new advance is made when the differences between the members vary in a determinate direction within the series. Such a regularly varying difference-series makes possible for reflection the formation of a law or rule by which the place of the subject-matters in the series can be determined. And the task can be set of finding new members in the series, intermediate members between those already given.

If the difference between members that have their determinate place in a series is constantly the same, member for member, we get an identically varying difference-series. The concept of identity begins to dawn, provisionally as identity of the difference between qualitatively varying subject-matters. Such series can be formed of moments, numbers, degrees and places. Every moment, with its peculiar lived experience, has a determinate place in the series of moments, and every moment stands in the same relation to the preceding and to the following moment. So too with the numbers. Every number has from the first, as Poincaré has expressed it, its own individuality, is qualitatively different from other numbers, not merely by its place in the number-series. That one can by different roads come to the same individual number is a discovery that presupposes a higher stage than the one we here stand at. Goethe held that mathematics moves in nothing but identities, because 2 + 2 = 4, but also 4 = 2 + 2, so that we get nothing other than what we had beforehand. But the different roads by which one can come to a number (e.g. by counting to 4 or by adding 2 and 2 together) are different operations, of which one discovers only afterwards that they lead to the same result and presuppose the same principles. The number is provisionally an ordinal number, a numeral. Differences of degree too are immediately differences of quality, and yet they lie in a determinate order in a series whose members can vary in an identical manner. So it stands with places in space as well. Even if they are perhaps still regarded as qualitatively different, it is an identical varying by which one passes from place to place.

β. In the series so far spoken of, the members are qualitatively different, and it is merely a question of their mutual relation with respect to place in a series. It is a great advance when this place becomes quite determinate. Classification and comparison become possible thereby. But these operations can really be carried out only when series can be formed in which every member is comprised in the preceding one. This can again appear in two ways. In a progressive difference-series every member can, for example, be a part of the preceding. In a classification the species is a part of the genus, and the individual a part of the species. From this it can be inferred that when an individual *A* belongs to the species *B*, and the species *B* to the genus *C*, *A* also belongs to *C*. It is essentially on this that the Aristotelian doctrine of inference builds. In a partial identity-series the following member stands to the preceding as predicate to subject, or as consequent to ground. Here too classification can be used as an example, for that *A* belongs to the species *B* means that the predicates *B* designates belong to *A*. In both the series named, the inference takes place by the elimination of intermediate members.

When a series in which the members lie fast is symmetrical, that is, can be read backwards just as well as forwards, one can infer from it to the reversed series.

γ. Progressive difference-series, partial identity-series and symmetrical series may be called rational series, inasmuch as they make inference possible. But the qualitative difference of the members is not abolished by their being capable of being seen as parts or wholes, or as subjects and predicates in relation to one another. The most perfect kind of inference becomes possible only when not merely the relation between the members is constantly identical, but the members themselves are so too, so that not merely members can be pushed out, but every member can be substituted for every other. We then get an absolute identity-series: $A = B = C = D \ldots$ Here is given, as Leibniz first rightly saw, the foundation — and at the same time the ideal — for all rational thinking. Such a series designates the complete opposite of the chaotic difference-series. The two series designate the ideal limits between which human cognition moves.

Step by step we thus come, by running through the stages the different series designate, up to the presuppositions on which strict science rests. But we see at the same time that the rational series are not the only possible ones, and that within the rational series a distinction must be made between those that make merely classification and comparison possible and those that make inference in the strictest sense possible. The different series form a scale, and it will perhaps be possible to find still more stages on this scale than have been named in the foregoing. The thought-work that is directed to the abolition of the chaotic relation between the subject-matters presents a whole series of degrees, until the purely rational stage is reached, where not merely the relation of the members but the members themselves are identical. This highest stage stands, however, always only as an ideal that can be reached only approximately and by abstracting from a quantity of actually present differences. Through ordering, classification and comparison, scientific consciousness works its way up towards its highest form, everywhere that the subject-matters make it possible.

Kant took account, in "Kritik der reinen Vernunft", only of the highest stage of science, that is, of the stage where it was possible to draw inferences on the basis of absolute identity. Classifying and comparing science he took no account of; that he reckoned to mere empiricism. And against the objection that classification and comparison nevertheless presuppose the validity of logical thinking no less than mathematical natural science does, he answers that the principles of pure logic are indeed negative criteria of truth, but not sufficient to ground the kind of science whose possibility he wishes to investigate.[7] But we do after all ascribe objective validity to our classifications and comparisons, just as much as to the natural laws proper, and there is therefore just as much ground for investigating the warrant for assuming the objective validity of purely logical principles as there is for investigating the objective validity of the mathematical principles and of the principle of causality. Salomon Maimon already saw this, in maintaining against Kant that the logical fundamental forms, affirmation and negation, have no significance if they are not presupposed valid for the subject-matters to which they are applied. Whereas Kant lets epistemology ("transcendental philosophy") presuppose formal logic as a discipline finished once and for all, Maimon maintains that epistemology must precede formal logic if the latter is to have real significance.[8] — Kant naturally had the right to limit the scope of his investigations, but he had not the right to think that he

[7] *Kritik der reinen Vernunft*[1], p. 151 f.
[8] *Versuch einer neuen Logik oder Theorie des Denkens*, Berlin 1794, p. 404–408.

had thereby also limited the scope of science.

b. From a new side the transition from elementary, prescientific consciousness to scientific consciousness comes forward when we bring out a presupposition on which all the kind of series-formation spoken of in the foregoing rests. We have in the foregoing taken account only of the struggle against the differences, and have seen how the degrees of likeness, both as regards relation and as regards members, step by step approach pure identity. But all difference and likeness, and thereby all series-formation, presupposes the possibility that subject-matters can be gathered, comprehended and ordered. Here the concept of synthesis comes forward as a presupposition that holds both for formal and for real science, both for classification and for the cognition of laws. Series-formation rests on determinate relations between the subject-matters being exhibited, and relation is thought only when the members between which the relation holds are the object of one and the same act of thought. Relation presupposes synthesis. But since synthesis, gathering-together, consists precisely in bringing subject-matters into some relation or other to one another, relation and synthesis are correlative concepts.[9]

Now the series spoken of in the foregoing are only typical examples of what can find application in many special cases. Many different classifications can be undertaken, many different comparative points of view adopted, many different laws found. But all these must again be brought into connection with one another, and hereby new series-formations will come forward. Syntheses of a higher order are then presupposed than in the single series taken each by itself. What matters then is that the scope of the synthesis becomes as great as possible. A striving towards an ideal maximum makes itself felt here. There will here be greater resistance to overcome than in the single series, since the difference between the series that can be formed in the different domains will be greater than the difference between members in the same series and in the same domain. The question becomes finally whether our thinking has at its disposal logical forms for complete gathering-together, or whether at a certain point more or less bold and more or less poetic analogies do not replace logical thinking. In the sixth chapter of the present treatise we shall return to this question. But before that boundary is reached where thinking passes over into poetry, there is still a domain in which scientific hypotheses

[9] In *Den menneskelige Tanke* p. 155 f. I set up synthesis as the first category, relation as the second. An acute reader has drawn my attention to the fact that they ought to have been set up as correlative concepts, like continuity and discontinuity, likeness and difference.

move with full right. The need for gathering-together will constantly make itself felt, not merely towards subject-matters, but also towards series. And by attempts at an ever more comprehensive synthesis light can be thrown on the single series and the single subject-matters, just as the single subject-matters are already illuminated by the place they occupy in the series of which they are members. Synthesis and analysis (and analogously: continuity and discontinuity, likeness and difference) stand in constant reciprocal action with one another, and science's rhythmical oscillations find their natural explanation in this fundamental law of human thought.

c. A third point of view for characterizing the difference between prescientific and scientific thinking is given in this, that the more science develops, the greater the differentiation of the different problems that takes place. In involuntary thinking no distinction is made between subject-matter, form and value. One does not make critically clear to oneself where the boundary lies between the immediately given (the subject-matter) and the thought-elaboration one applies in interpreting it. Nor is it recognized how even the subject-matter's first appearance for us is conditioned by our receptivity, which is never absolutely passive. Neither is value definitely separated from subject-matter and form. The point of view of value will under primitive conditions have been the predominant one, since subject-matters are noticed especially on account of their practical significance, and since the way in which the subject-matter is interpreted is also conditioned by its connection with practical interests. Bergson ought therefore to have reckoned the element of value among "les données immédiates"; he has attempted to eliminate the element of form and thinks that if it could be separated out, the element of subject-matter alone would remain. He is even inclined to think that the ground for the predominance of the element of form in science must be sought in practical considerations. But interest cooperates already in the choice of subject-matter. Naturally it cooperates also in the primitive interpretation. The teleological point of view rules in animism under all its forms, from fetishism right up into the most sublime theology. And in the primitive forms or intimations of what we now call categories it can be traced that practical motives have cooperated in their arising. Space means at first distance from an end striven after, time a relation between a wished-for future and an unsatisfying present, and the causal relation makes itself felt in the attempt to abolish these distances in space and time. In the course of science's development a constant striving is traceable to eliminate the element of value, to abstract from all interest with the exception of the intellectual interest, which both as motive and as effect is bound in the most intimate way to thought-work itself. Thereby other values do not necessarily fall away, but the concept of value can now become the object of a special problem, which among other things also discusses the relation between intellectual value and other values.

The distinction between the three elements marks an important advance. It has not yet been strictly carried through in science, and still less in ordinary consciousness. This distinction naturally solves no problems. On the contrary, it is the condition for problems being able to be set sharply and determinately, and the very clash between the three elements sets in the end the greatest problem of all. — For the rest, this point of view, drawn from differentiation, leads back to the preceding points of view, inasmuch as with increasing differentiation increasing demands are made on synthesis, if the unity of the life of the spirit is to be preserved, and inasmuch as differentiation occasions new relations.

5. Psychology's independence over against epistemology follows from the fact that scientific thinking cannot emancipate itself from the general laws of the life of consciousness. (Cf. my *Psykologi*[6] V B, 11.) Both discovery and the conducting of proof are due to psychical work. They must be psychologically possible. One cannot deduce epistemology from psychology, but one can investigate what psychological conditions scientific cognition presupposes, and then investigate whether these conditions are present. Psychology's position over against science is quite analogous to its position over against art and religion. It asks how science is possible, just as it asks how art and religion are possible. These different spiritual forms of life grow out of involuntary mental life and, so long as the conditions for their subsistence are present, do not lose their connection with it, in spite of the peculiarity with which they appear in their higher development.

When epistemology is presented synthetically, the connection with psychology shows itself most plainly. Psychology then appears as an introduction, as a kind of "Phänomenologie des Geistes". In investigating involuntary thought-work we become aware of that relation between subject-matter and interpretation which plays a decisive part in scientific thinking. The forms under which subject-matters are worked up by involuntary thought-activity must be ground down, widened or narrowed, if the goal science sets itself — namely to win thoughts that can subsist in spite of the changing of subject-matters — is to be reached.

Kant proceeds synthetically in "Kritik der reinen Vernunft". His "metaphysical" deduction in the "transcendental aesthetic" and his "subjective" deduction in the "transcendental analytic" are in reality psychological analyses, by which the concept of form is separated out from the concepts of subject-matter and value. In my book "Den menneskelige Tanke" I also proceed synthetically, only with greater weight on the psychological and historical foundation. I seek to show how the transition from intuiting to judging already contains the germ of scientific thinking, and how the latter, through the history of science, awakens to an ever clearer consciousness of itself. From this followed the division of my presentation into three sections: Functions, Categories and Problems.

But in an analytic presentation of epistemology too it will constantly show itself that there is a necessary connection between epistemology and psychology. One then starts out from actually existing science and asks what presuppositions it rests on and what forms of thought it works with. And thereafter one cannot avoid asking how far these presuppositions and forms are actually present in the life of consciousness, and how they stand to its involuntary thought-work. This road Kant took in the "Prolegomena". The road leads here from the fundamental propositions through the categories back to the psychical functions, which it is psychology's business to exhibit and to describe. This triad emerges clearly in more recent epistemological works that proceed analytically, as the "Prolegomena" does. Meyerson (Identité et Réalité, 1908) comes, through analysis of the presuppositions of modern exact natural science, back to the opposition between identity and succession. Causality in the strict sense contains, on his conception, two-sided equivalence, possibility of substitution, and therefore identity. But there can be lawfulness (*légalité*) without two-sided equivalence; here substitution is not possible, and therefore Meyerson will not use the expression causality here. Carnot's principle (entropy) shows the significance of this difference. Meyerson finds through the history of science a constant, irrepressible tendency in the human spirit to form concepts of something that subsists in spite of all succession. But when now time cannot be eliminated — which Carnot's principle plainly proves by exhibiting a limit to the substitution of states in nature — then we stand at the opposition between identity and succession, and at the opposition, connected with it, between quantity and quality, which points back in the end to the relation between thinking and sensation, which it is psychology's task to investigate. So here too we come back from fundamental propositions through categories to psychical functions. In Cassirer (Substanz und Funktion, 1910) as well, the difference between functions, categories and problems emerges clearly, in the reverse order from that which I, on account of a synthetic presentation, follow in my book. Cassirer's analysis of the presuppositions of science leads him to impress the significance of the concept of relation, which for him, as for so many more recent epistemologists, becomes the most important category. And thereby the task is in the end set for psychology of investigating relation, the part played by determinateness by relation within mental life. Wrongly does Cassirer think that this investigation has hitherto been altogether neglected. But the main thing in this connection is that there is a close connection between epistemology and psychology. Indeed, Cassirer even ends his book with the declaration that in the end the opposition between them falls away, since "psychology itself gives a start (Ansatz) towards the problems that must seek their advancing solution in logic and in logic's application to science". In Henri Poincaré's epistemological treatises too we find the triad spoken of. His investigations of the theories of modern mathematics and physics lead him back to the opposition between continuity and discontinuity and their constant clash within the development of thought and of science. And this opposition it is psychology's task to investigate, since it hangs together with the opposition between intuiting and thinking, an opposition that must constantly assert itself anew. Poincaré therefore denies (see especially "Dernières Pensées", p. 139, 158) that there is any epistemology independent of all psychology: such a thing is just as impossible as that there should be science without scientists!

If involuntary thought-life did not prepare the way for scientific thought-life, science would lose its natural ground. Science grows up out of involuntary thought-life. There is accordingly a whole group of categories, the fundamental categories, with which involuntary thought-life already operates: synthesis and relation, continuity and discontinuity, likeness and difference are forms that no member of consciousness can do without. They yield a definition of what it is "to think" in the widest sense of the word, the sense in which Descartes, by his own explanation, took the word when he said "je pense, donc je suis". The other groups are specially formed applications of the fundamental categories. In classification and in logical and mathematical thinking the formal categories (identity and rationality) display themselves. In natural science, psychology and sociology one works with the real categories (causality, totality, development). In economics, aesthetics, ethics and the philosophy of religion, value-concepts lie at the base. A rational classification of the sciences is one and the same thing with a systematic doctrine of categories.

The formal categories came to consciousness already in antiquity. Here lies the great deed of Greek thinking. In more recent times it cost a struggle before the real categories could be seen in their characteristic difference from the formal. Here lies the great contribution of Hume and Kant. And it has cost — and still costs — labour to separate the value-concepts out from the other categories and yet to see them in their connection with them. The history of the problems and the history of the categories stand in close connection with one another.

Some examples will further illustrate both the relation between the different groups of categories and the relation between psychology and epistemology.

It is by analysis of the states given in psychological experience that epistemology is prepared. In the psychical states there is given something other and something more than epistemology needs. It is the tasks that determine what is selected within the sum-total of possibilities presented. There appear, for example, in psychological experience different degrees or kinds of relation of likeness. In recognition, association and comparison alike the relation of likeness displays itself, partly as coincidence, partly as composite likeness (coincidence + difference), partly as qualitative likeness, partly as likeness of relation. These likenesses acquire epistemological significance in that they make possible the formation of series — the series lying fast, in which the members are ordered according to decreasing likeness or increasing difference, and by which classification and comparative science become possible. But logic and mathematics require coincidence in its perfect, ideal form as absolute identity, because only thereby do strict inference and calculation become possible. Identity is then defined as a limiting case of likeness. It is an analytic definition that we here apply, just as when in geometry we go from the body ("the block") to the surface, from the surface to the line ("the edge"), and from the line to the point. The beginning is psychological (or lies in a domain common to psychology and epistemology), but from this starting-point the epistemological point of view develops successively to independence. This development coincides in part with a development from practice to theory. In practice we can often be content with an estimate, an apprehension of likeness of a kind lying between coincidence and likeness of relation. But strict theory requires identity as the foundation for inference and measurement. It is such an identity that Plato means by what he calls "the Idea of likeness". The Platonic Ideas are both general concepts and fundamental concepts; Plato felt no need to distinguish sharply between classification and exact thinking, between descriptive and formal science. Just as for Plato the highest degree of likeness is an "Idea", to which in all given cases we have only approximations, so according to Kant pure space is an "Idea", a rule by which phenomena can be ordered and judged, but which is not itself a phenomenon. The strife between the different epistemological tendencies really turns on how far it is necessary to go to the limit — to identity as the ideal of likeness. All are indeed agreed that absolute identity cannot be exhibited in any observation whatever. It is merely a question whether one will go at once to the practical application of science, or whether one will first present science in the strict form it acquires by one's going to the limit, and thereafter descend to the cases possible in experience and practice. It is a relation between whole and part, or between a series of increasing degrees of likeness and this series' last member, that here lies at the base. The different definitions that have been given of identity have this in common, that they all presuppose likeness and difference as known, as immediately presented as soon as reflection has begun to work, and that within the possibilities the relation of likeness and difference contains one selects the one that can become fruitful for thought. When Leibniz defines identity by possibility of substitution, that means, after all, that he keeps to the degree of likeness that makes it possible to put the one in place of the other. So too in a famous mathematician's definition of identity as the difference that is smaller than any given difference; here a series of decreasing differences is simply laid at the base. The smallest possible difference is an "Idea", just as the greatest possible likeness is. A third definition (Bolzano's), which explains identity as a thing's likeness with itself, leads back to Leibniz's definition, since it is such likeness that conditions substitution. Or else it leads back to the concept of the smallest possible difference; for between a thing and itself there is as a rule less difference than between a thing and other things. And since no thing appears in quite the same way in different connections, identity becomes on this definition too an "Idea".

It is not a matter of indifference to psychology whether one defines identity by likeness or likeness by identity. In the latter case the definition becomes synthetic; one apprehends identity as a simple concept and likeness as identity + difference. But against this psychology, and at bottom all science of experience, must protest. For there are likenesses that cannot be resolved in this way, as for example likenesses between two colours (orange and yellow, blue and violet). Not merely qualitative likeness, but likeness of relation (analogy) too, cannot without more ado be resolved in this way. —

Another example of a transition from psychology to epistemology through analysis is to be had in the relation between the criterion of reality and the concept of cause. Psychology must for its own part set itself the question how men go about it when they come to doubt the validity of their observations or representations, that is, the possibility of holding them fast and acting according to them in all cases. Now experience shows that in cases of doubt we involuntarily set about testing whether the single observations and representations agree with one another and with the new observations that can be made. Only where an unavoidable connection shows itself between all the different observations that stand at our disposal do we rely on it that we stand before a "reality". But such an unavoidable connection is precisely the most elementary form of the concept of cause. Science does not remain at this elementary form, but seeks, everywhere it is possible, to make the connection absolutely rational, so that the members become two-sidedly equivalent. In the ideal concept of causality, rationality has the preponderance over the spatial or temporal relation between the members. — Psychologically it is asked how we go about it in cases of doubt; epistemologically it is asked with what right we infer from past or present to future, or from present to past.

6. It is no objection to the determination here given of the relation between psychology and epistemology that psychology itself, like any other science, presupposes an epistemological foundation. Epistemology, which investigates the foundation of all science, naturally investigates psychology's too. Psychology's scientific character rests on the fact that in the domain it cultivates series can be formed that can acquire scientific significance. It is provisionally series with members lying fast that will be able to be formed in the psychological domain; absolute identity-series, and thereby the possibility of exhibiting exact relations of equivalence, still lie far off.

There is no contradiction in epistemology, which has itself historically developed out of psychology, later turning against its own origin — among other things in order to understand its own coming to be — any more than there is any contradiction at all in the history of science becoming an object for science. It is not, after all, epistemology's task to produce the special sciences. It takes them as given, although they do rest on presuppositions that it is epistemology's task to investigate.

The difficulty often found here will perhaps disappear when one considers that epistemology itself rests on certain presuppositions as soon as it is to be a science. There can accordingly be an epistemology of epistemology, an epistemology in the second power. When epistemology asserts, for example, that all cognition presents a relation between a subject (S) and an object (O), this relation itself and its members (S and O) can be taken up again and considered as an epistemological problem. This is at bottom what happens when epistemologists criticize one another. When one does not, like KANT, think that epistemological work can soon be brought to a close, one must concede that there is here the possibility of an infinite series. One cannot, once and for all, break the sting of scepticism. It is a fundamental peculiarity of the life of the spirit that preceding spiritual work can become the object of new spiritual work. The subject can make itself into an object. If the sign { expresses a relation of objectification, we can get an infinite series:

$$S_1\{S_2\{S_3\{S_4 \ldots$$

In this series O has been entirely abstracted from. In O changes can occur that do not depend on S alone, just as in S too changes can occur that do not depend on O alone. The relations will accordingly in reality be more complicated than in the series set out. But this one is sufficient to show what matters here. — Cf. on the whole question here touched on: "Den menneskelige Tanke" p. 297–305.

7. As for all concepts drawn immediately from intuition (sensation, memory or imagination), there holds for the categories the rule that there is an inverse relation between content and extension. The more comprehensive categories are the fundamental ones (synthesis and relation, continuity and discontinuity, likeness and difference), since they hold for all conscious life, for all sensation and thinking, and for invalid thinking just as much as for valid thinking. A narrower class of categories is the formal ones (identity, analogy, negation, rationality), which display themselves when determinate demands are made on the formation of concepts, judgments and inferences. When it is asked how these formal categories are applied to the qualitative changes that observation exhibits, the real categories (causality, totality and development) come forward. And when it is asked in particular how far conditions are present for a certain whole's arising, subsisting and developing through time, the ideal categories, the value-concepts, come forward, since all value presupposes a relation between a whole and its conditions.

The categories can therefore be ordered in a progressive difference-series, thus — when we keep to the most important categories within each single group —

Synthesis > Identity > Rationality > Causality > Totality > Value.

The following members in the series constantly presuppose the preceding, are in a way contained in them, are special formations or applications of them. Historically the categories have come forward sporadically, according to the need there was to apply them, the tasks that had to be solved. The different sciences, each working with its own categories, also arise sporadically, even if Comte is right that in the order in which they attain a strict, positive character a certain order can be traced. On the whole there has been a tendency to work with totalities and values before working with the preceding categories. The fundamental categories are first brought out with clarity by Leibniz and Kant. Precisely because they are most intimately bound up with, or are one with, our spirit's own manner of working, it is understandable that they are discovered late. The logical order is in any case a different one from the historical. As with the classification of the sciences proposed by Comte, we can order the categories so that one advances from the most necessary, and therefore most comprehensive, to the more special.

It shall be investigated in a later connection whether it is not possible to widen at once a concept's content and its extension. The logical rule spoken of, about the inverse relation between content and extension, holds in any case for the categories, since these are drawn immediately from the course of the movement of thought in conscious life and especially in science. It is impossible to derive the other, more special groups of categories from the fundamental categories. The law of synthesis could thus hold even if rationality and causality could find no application; it holds, for example, for a chaotic difference-series too. Causality could hold even if we could find no application for concepts like totality and value. Hegel's magnificent attempt at a purely deductive development of the doctrine of categories out of the simplest and most comprehensive fundamental concepts foundered on the necessity of constantly taking up new elements and premises from experience. Conversely, it is impossible to make the most concrete categories, the value-concepts, into the most universal, as is attempted in all theological or theologizing views. Both alike lead only to a play with analogies, without the possibility of exhibiting both the warrant of these analogies and their limitation.

But it follows already from the position of the concept of synthesis as the first member in the series of categories that there is a type common to all categories and groups of categories. If synthesis is the common form for all thought-activity, in whatever domain it is exercised, all thoughts must also present a more or less pronounced character of wholeness. And if it is right that all fundamental definitions must be analytic, then it lies in this that we always begin with wholes, and that thought-work consists in finding parts or elements within such wholes. Thus identity is a special case of likeness, the surface a part of the block, and so forth. Apart from the concept of totality's special place in the series of categories, it will be to be presumed that it is anticipated in the preceding members of the series and lies at the base, in more special form, in the following members. It is from this side that the present treatise takes up anew the investigation of the categories. The ordering of them presented earlier in "Den menneskelige Tanke" will hereby receive a closer elucidation.

8. Historically, as already indicated, the concept of totality makes itself felt very early, on account of the tendency to begin with the most concrete concepts, of which one cannot at once see that they are at the same time the most complicated — or perhaps rather because the different groups of categories had not yet separated themselves out from one another, but formed a single whole or were regarded as identical with one another.

The ancient picture of the world stood before thought as a completed whole, within which every being, every quality and every value had its determinate place. This presupposition announces itself already in the Pre-Socratics' question: "What is it that is?" A property of existence is sought which holds for the whole and all its parts, in equal degree *in toto et in parte*. But hereby there arose at the same time a problem that stretched through the whole history of thought. For when that fundamental property is to be the same everywhere, whence then comes the difference between beings, between qualities and between values? The differences cannot be derived from pure identity, and there arises therefore in strict logicians, like Parmenides, a tendency to deny the reality of the differences or to relegate them to "the opinions of mortals". There appears here an opposition between unity and whole, between rationality and totality, an opposition that in more recent times appears as an opposition between formal and real science. In ARISTOTLE's doctrine that science has to do only with the universal, but that everything actual is individual and therefore bears the stamp of totality, the problem we here stand at appears in a characteristic manner — and for all time.

In the Renaissance the concept of the whole presses forward into the first rank, and attempts are made to overcome the ancient opposition between thought and totality by interpreting PLATO's "Ideas" as laws for the real connection of phenomena. The true Ideas, says BRUNO, are not concepts of things or properties in their abstract essence, but concepts of laws that express the connection of different parts in a whole — for example not concepts of head, lung, heart and so forth each by itself, but of the connection between all these things, whereby they come to constitute a whole.[10] The train of thought thus intimated is continued in SPINOZA in his doctrine of individual wholes that can subsist even if the parts change and the single processes within the whole alter direction, provided only that the law of the connection continues to be the same, — and of individualities of different degrees, forming a graded series, until at last the whole of nature stands before us as one great individuality, whose single parts vary without the great individual as a whole undergoing any change.

The cognition of laws that modern science aims at concerns the relation between the phenomena that experience offers. One wants to get away from the barren general concepts to which one can climb up from the single observations, but from which one cannot climb down again to the particulars that observation establishes. This is the momentous change of task to which modern natural science owes its coming to be, and which found its expression in GALILEI's doctrine of method in the demand for cooperation and mutual supplementation between deduction and induction. But along this road one cannot come to the thought of such a completed whole as antiquity held fast to, — only to a great, unsurveyable connection that can always be extended. BRUNO already intimates that neither the concept of part nor the concept of whole can strictly speaking find application to existence.[11] SPINOZA attempts, as we shall come back to in a later

[10] *"Den nyere Filosofis Historie"*, I, p. 126 (BRUNO). Cf. p. 194–196 (BACON).

[11] L'universo è tutto in tutto (se pur in modo alchuno se può dir totalità dove non è parte ne fine). *Opere italiane*, ed Lagarde, p. 315.

connection (Chap. V), to unite the cognition of laws and the conception of the whole in one concept. But Hobbes declared definitely that about the world as a whole one can raise only a few questions — and none of these questions can be answered.[12] The train of thought Hobbes intimates is completed in Kant's doctrine that concepts of absolute totality, "Ideas" as he calls them, cannot serve as a foundation for scientific cognition. And among the "Ideas" he reckons, among other things, the concept of the world; if this concept is to be capable of being formed, it will presuppose an infinite and yet completed synthesis.

The fundamental categories already make the demand of everything that is to be capable of being cognized, that it must be able to be comprehended together with something else and placed in relation to it as continuous or discontinuous with it, and as like or different from it. That is why Hobbes is right that one cannot get an answer to what one asks concerning the world as a whole. But there are all the more questions to raise in return, with prospect of an answer, concerning the limited individualities or totalities that experience shows us. Since every limited whole stands in relation to other wholes or to elements different from itself, it becomes possible here to apply our categories.

A limited whole cannot have the ground of its arising, subsisting and passing away in itself. It is a member in a series, presupposes preceding members in this series which have prepared it and from which it successively separates itself off; it must at every moment come into being anew, since its subsistence depends on determinate conditions; and in so far as it is the cause of changes outside it, it will always show itself to work together with a manifold of other causes, without which its peculiar intervention would not be possible or would lead to no result.

But a problem arises not merely from a whole's relation to what is different from it; in the relation between the whole as such and the elements contained in it there lies a problem too, one that might perhaps tempt us to deny the concept of the whole any epistemological warrant at all. There stand here in any case different tendencies over against one another, corresponding to the opposition between deduction and induction and conditioned by the difficulty of applying here the reciprocal action between the two methods that Galilei demanded.

[12] De mundo, ut uno multarum partium aggregato, qvæ qværi possunt, paucissima sunt, quæ determinari, nulla. *De Corpore*, Cap. 26, 1.

From the one side the whole weight is laid on the elements that can be distinguished within a given whole, that are of more or less far-reaching significance for its existence, and that can be found again outside it either in the same or in analogous forms. — From the other side the whole weight is laid on the peculiarity with which the whole appears in experience, on the inner connection between its elements so long as they are members of it, on the concentrated force it is able to apply in quite determinate directions and without which it could not subsist. By reason of this inner connection and this concentrated action it presents properties that none of its elements has by itself, and it thereby comes forward as different in kind from these. In every attempt to write the history of a mountain, a river, a forest or a planet this difference comes forward. But it is especially in the biological, the psychological and the sociological domain that an opposition, and with it a problem, makes itself felt here. It is the opposition between the tendencies that can be designated respectively as mechanicism and vitalism, as associationism and spiritualism, as individualism and socialism.

It has been sought to find the difference between natural science and historical science (or, as some would rather say, cultural science) in this, that the former aims only at finding general laws for the cooperation of elements, whereas the latter has to do with individual wholes in their peculiarity. Over against such a conception too a closer investigation of the concept of totality will be of importance. If this concept, like all other categories, bears the peculiar stamp of human thought, it will not be capable of being wholly denied in any domain of cognition whatever.

In some famous lines Goethe has struck the note of an inquirer's mood in face of the concept of the whole:

> Freuet euch des wahren Scheins,
> freuet euch des ernsten Spieles!
> Kein Lebendiges ist eins,
> immer ist es vieles.

The whole has the character of semblance and of play when it is taken as fully finished at any stage whatever, and when one in a mystical manner ascribes to it a force of self-preservation and development, quite apart from the reciprocal action between inner elements and with outer conditions. But there is truth and earnest in the semblance and in the play, because the connection of the whole conditions the character of the elements, no less than they determine the character of the whole, and because outer conditions gain influence only through the cooperation of the peculiarity with which the actually given whole appears. Moreover, thought-activity itself, in whatever domain it moves, is the expression of a whole that seeks to extend itself under an ever more intimate joining together of its elements. — From these different points of view we shall in what follows consider the concept of totality.

Chapter II

The Concept of Totality and the Fundamental Categories

9. Without involuntary whole-formation, reflection, which begins with analysis, would not be possible. A concrete intuition, the fruit of a synthesis that has taken place without being an object of consciousness, forms the foundation of all cognition.

Neither an absolute single and unique thing, nor an absolute chaos, could form a foundation for reflection. The fact that analysing reflection is possible — this fact, which is the foundation of all science — proves that the primary psychical process consists in a whole-formation.

Reflection directs itself partly to the relation between the changing wholes (in the domains of sensation, memory and imagination), partly to the relation between the different parts or properties of the single whole. Every subject-matter is a little world, but this little world shows itself also to be part of a greater world, or to stand in relation to other little worlds. Every subject-matter has, as has been said, rents or fringes at its edge, which betray that it has been taken out of a close connection with other subject-matters. Step by step it is discovered that every determination — every determination of time or place, for example, every establishing of a property — at bottom presupposes a far greater whole than the one reflection had begun with.

The single sensation of sense is already determined, both as regards degree of strength and as regards quality, by the whole connection within which it occurs. In recognition a connection with the after-effects of earlier sensations also makes itself felt. Processes take place here for which language has no peculiar names, but which present analogies to what at more developed stages of consciousness appears in the relation between given subject-matters, their interpretation, and the inferences drawn from them — an analogy that must not be confused with identity. In memory and imagination whole-images are formed in virtue of the law of wholeness underlying all association of representations, which testifies to a need and a capacity to fill out and continue given representations. As already mentioned (§ 3), concrete intuition is held fast as long as possible, and displacements and articulations can take place within it without proceeding to the formation of judgments. Besides the examples I have adduced in earlier presentations, I shall here adduce an example of articulation within a memory-image. My memory-image of a vote on the award of a scholarship contained, among other things, the image of a table on which voting slips were lying. But when the memory-image came up again on a later occasion, it was not voting slips that lay on the table, but letters. I then naturally came to compare the two memory-images with what I knew by another road, and passed the judgment that the latter was right: for the voting slips were sent in in sealed envelopes. The memory-image had accordingly involuntarily undergone a displacement in the right direction, and it was only the relation, the contradiction between the two memory-images, that made the judgment necessary.

An example of articulation already touched on arises from the tendency to make a surface's figure and its outline into one. For the figure cannot be localized; it concerns the whole surface. The surface-figure is what is immediately experienced; but the outline is both easier to describe and of more practical significance. That the outline presses forward in relation to the surface-figure is an intimation of an analysis that can later take place, but that is not absolutely necessary, since the displacement can go on within involuntary intuiting. As Edgar Rubin, who has clearly elucidated this point, remarks, we stand here at an expression of a peculiarity of the psychical that makes itself felt in other domains too. Just as little as the surface-figure can be localized anywhere in space within the given whole, because it belongs to the whole itself, just as little can experiences of rhythm, movement and duration be referred to any single point of time within the whole experience.[1] Beginning analysis runs, as the Eleatics already saw, into great difficulties here. The whole itself stands in danger when analysis begins, since it is

[1] *Synsoplevede Figurer*, p. 179.

incommensurable with every one of the particulars analysis can point out. Apparently uncompounded sensations of sense are little wholes that do not receive their due through analysis. Thus the sensation of roughness is a whole made of elementary sensations of small unevennesses that are not noticed each by itself; no single sensation of such an unevenness would give the sensation of the whole. — On account of this peculiarity of consciousness the need to hold fast to intuition as long as possible is understood.

Of a peculiar kind of displacement we have an example in the images of light and colour that tones can call forth — what have been called musical photisms, where the mood called forth by tones, and especially by timbre, finds immediate expression in visions of figure or of colour. It is an "instinctive translation into visual expressions" of what has affected the sense of hearing.[2] When elements of feeling as a rule make themselves felt here, that is an example of concrete and practical intuition passing over into one another, since in the involuntary whole-formations in the domains of sensation, memory and imagination there will always, in greater or less degree, be need and feeling at work.

In the involuntary need for whole-formation the representation of the self has its ultimate ground. It is due to a whole-image formed of the most constant sensations, memories, moods and endeavours, a whole-image that will always be articulated in a determinate direction according to the ruling interest. Especially such states as stand before us as causes of changes in us or outside us exercise their influence here. There works here what GADELIUS has called a motor feeling of wholeness.[3] To express consciousness of ourselves (or of our self) analysis belongs. And analysis has here, as everywhere, its difficulties. Here too we keep as long as possible to what immediate and actual intuiting yields. Hence comes the mystical element that often makes itself felt in the representation of ourselves. Here as everywhere mysticism is warranted when it is the expression of the inexhaustible in the given. Our representation of ourselves is for the most part a practical intuition. Analysis can bring forward only a single trait at a time; and when are all the traits brought forward, and what guarantee can we have that they have all been found? Our self-knowledge is therefore always imperfect, quite apart from the fact that, so long as life lasts, new lived experiences can come and with them possibly fundamental

[2] TH. FLOURNOY: *Des phénomènes de Synopsie*, p. 103.
[3] BROR GADELIUS: *Tro och öfvertro*, I, p. 158.

alterations in the total connection out of which the representation of the self is to emerge by analysis. Here too it is understood why we cling to intuition as long as possible. We appeal especially to our conscience when we experience, or think we experience, the insufficiency of analysis.

When concrete and practical intuition are no longer able to contain the subject-matters, either on account of their manifoldness or on account of their mutual oppositions, there is a need for, and perhaps also a possibility of, a new kind of whole-formation, namely the formation of judgment. Referring on this to my earlier presentations ("Det psykologiske Grundlag for logiske Domme" and "Den menneskelige Tanke"), I shall here merely make some remarks on a theory put forward by Heinrich Maier in his work "Psychologie des emotionalen Denkens", already mentioned above. With very much in this work I can express my complete agreement. The difference between Maier's presentation and mine often rests only on a difference of terminology. But on two points I cannot agree with him. On the one hand he extends, as already mentioned above, the concept of judgment so as to hold for all observations, memories and imaginings, because they are not "indifferent" to the opposition between true and false. On the other hand he maintains that "emotional thinking" does not lead to cognition, but only to "presentation" of what is the object of our feeling and our need.

The first point I have already touched on above (§ 3), where I have sought to show that Maier wrongly transfers the opposition between true and false into the primary intuiting. It plays no part there, because a secure repose prevails in what is sensed or represented, a security that knows no doubt and therefore does not distinguish, and cannot distinguish, between true and false. Adam and Eve were not indifferent to the difference between good and evil before they ate of the tree of knowledge; they did not know this difference at all. And that immediate security cannot be said to be indifferent to a difference that does not exist for it.

I return here to this point in order to draw attention to the fact that Maier has two marks by which he decides whether a judging makes itself felt or not. The one is the already mentioned opposition between true and false. The other consists in a transition being made from an independent member within an object to another independent member. He acknowledges that a relation-positing (Beziehungstätigkeit) can make itself felt which is not yet a judging, because the members do not stand as independent over against one another.[4] This corresponds to what I call articulated or articulating intuiting, a phenomenon which I do not think MAIER allows to come fully into its own.

It is in my view only this second mark, the transition from an independent member to another independent member, that marks the judgment in contrast to the intuition. And it presupposes analysis; for in immediate intuiting the members do not stand as independent. They are not yet separated out from the involuntarily formed whole. Such an analysis and transition can moreover very well take place without the difference between true and false making itself felt. A judgment can be passed just as involuntarily as an intuition can be formed. Immediate security can prevail in our judgments just as well as in our concrete and practical intuitions. It is only when doubt — that is, a certain special kind of judgment — arises that that difference can come forward.

MAIER's two marks come into conflict with one another, since there are judgments in which the opposition between true and false does not make itself felt, and since it is nevertheless the presence of this opposition that is supposed to make it necessary to extend the concept of judgment itself to activities of consciousness that do not consist in a transition from one independent member to another.

As regards the second point, I cannot see otherwise than that MAIER, in spite of the merit he has earned by his characterization of thinking determined by feeling and will, has not let "emotional thinking" come fully into its own as a function that is at bottom of quite the same kind as "cognitive thinking", although it works under peculiar psychical conditions. One would after all also expect that thought-activity works essentially in the same way, whether it goes on predominantly on the basis of sensations, observations and representations, or predominantly on the basis of need, pleasure and displeasure.

MAIER here uses the first of the two marks of a judgment mentioned above. Emotional thinking, he says (p. 140), does not aim at being true, and it is therefore no judgment. But is it now right that emotional thinking does not aim at being true? According to MAIER himself its significance consists in its bringing the object of our pleasure and displeasure, or of our need, clearly and distinctly before us. Therefore MAIER also speaks of self-evidence in volitional acts of thought, a self-evidence consisting in the tendency of desire coming to adequate expression in representations of desire. And it is especially when a doubt has first made itself felt that this self-evidence plainly appears (p. 42). I

[4] *Psychologie des emotionalen Denkens*, p. 123.

cannot see otherwise than that the concept of truth puts its head out in such expressions as "adequate" and "self-evidence". And the relation is here in reality the same as with "cognitive" functions, which also make themselves felt most plainly after a doubt. What I sense, remember or imagine can indeed become more or less "adequately" or "evidently" conscious to me, that is, my representation of it can more or less completely cover the content of the sensation, the memory or the imagination, can be the fruit of a more or less complete analysis of the immediate intuitions. And so it can also be asked, of the representation that concerns the goal of a need or an impulse, whether it too is true, that is, completely expresses the real direction of the need or the impulse. It is still plainer in the case of a resolution: is what I have set before myself also what I innermost will? Is there truth in my willing, that is, does it cover my real striving? Is it not an illusory goal I have set myself, that is, a goal I do not at bottom aspire to? In the consciousness of a willing, illusions are just as possible as in observations of sense. And there is an analogy between purpose and observation: in the moment of decision I see myself acting in a future moment, anticipate (perceive) a future situation. And it must be added that even if the purpose is carried out, it is not thereby said that it was the expression of what was in reality the innermost striving.

There is, as Danish philosophers especially (Poul Møller, Kierkegaard) have impressed, a personal truth, which consists in the goals we consciously set for our striving agreeing also with the innermost essence of our personality.

Instead of distinguishing between cognitive and emotional thinking, one should therefore distinguish between two kinds of subject-matter: those that appear as mere, essentially uninterested observations, and those that occur with the character of need and striving. Reflection works in the same way towards both kinds. In both cases we begin with involuntary security, without being "indifferent to true and false", but with the possibility of being entangled in illusions. And in both cases room can arise for a judging in the sense of a transition from member to member, and thereafter perhaps for a critical testing that abolishes our primary security. The representations that are interwoven — now more as conditions, now more as products — in our need and striving,[5] our impulses

[5] Maier attaches great importance to the difference between involuntary *and* voluntary willing (558 ff.; 574); but he sets the boundary at another place than I do, since he reckons impulse and wish to involuntary willing and makes deliberation the presupposition of voluntary willing. Spontaneous and reflex movements he does not reckon as psychical functions, and instinct he reckons to impulse, maintaining that in it too

and wishes, our intentions, purposes and resolutions, can just as well become members in judgments and objects of criticism as the representations interwoven in our observations, memories and imaginings. It is a merit of MAIER's that he has so energetically drawn attention to them. But he has not thereby exhibited any wholly new kind of thinking. The fundamental categories occur in both kinds, and when criticism has awakened the other groups of categories will also be able to make themselves felt. In the discussion of the value-categories (Chap. V) we shall come back to this question.

10. Reflection presupposes an immediately given in the form of intuiting (concrete or practical intuition), and this immediately given stands as a whole that can now become an object of analysis. In such a whole a manifold of elements shows itself to be connected in a peculiar way, and we therefore find the law of synthesis holding already here.

Against the warrant for finding synthesis in the immediately given, HERMANN COHEN has raised the objection that synthesis presupposes a given manifold that can be gathered together, and he has held that KANT, in applying the concept of synthesis, makes himself guilty of LOCKE's and HUME's error, namely that of regarding the originally given as a chaos of sensations of sense.[6] In a somewhat different form the same objection appears when BERGSON[7] maintains that the immediately given cannot be expressed by a concept that contains an opposition to another concept — and the concept of synthesis does after all contain an opposition to the concept of manifold.

Against this objection it must be conceded that the concept of synthesis has been carried over by analogy from more differentiated psychical functions to the more primary ones. Language has, as already remarked, formed no expressions for the most primitive psychical functions. Synthesis appears most plainly where the task is set of connecting and ordering actual and plainly presented differences. In consciously set tasks we have both the preceding manifold and the connection won by psychical work before us as objects of consciousness, and perhaps different intermediate stages of advancing ordering and wholeness. By the road of analogy we apply the concept of synthesis, and with it the concept of psychical work, even where no preceding manifold can be observed

there is always a representation of what is aimed at. I do not believe observation confirms this division. A need can make itself felt and find vent in involuntary action without any representation of a goal being at work.

[6] H. COHEN: *Logik der reinen Erkenntniss* (1902), p. 24.

[7] BERGSON: *Introduction à la Métaphysique* (Revue de Métaphysique et de Morale, 1903), p. 15–17.

and no intermediate stages in its gathering-together apprehended — as in immediate intuiting, that is, in the emerging of images of sense, of memory and of imagination with that character of wholeness which is the condition of reflection's analysis. It is both continuity and analogy that lie at the base here. For there can be all possible degrees in the speed with which the gathering work is exercised, and experience shows that the shorter the physiological time is, the more consciousness is replaced by unconsciousness. And everything indicates that even what psychologically appears as single and completed is, physiologically regarded, represented by a manifold of processes. We have therefore no right to think that psychical work (to which a physiological work is naturally always assumed to correspond) ever falls away entirely, so long as consciousness subsists. There are all possible degrees between potential and actual psychical energy.[8]

For immediate intuiting (concrete and practical intuition) there are no problems. They come only with reflection. If it is a matter of getting into a problemless state, all that is needed is to keep reflection out. We all have a need for that, and it is warranted too, when we are to be wholly and fully absorbed in a lived experience. The value of imagination and of art presupposes such an absorption; so it stands with the conditions of life we are absorbed in, and with devotion to great ideas and endeavours. But life constantly demands new psychical work, since synthesis runs into new resistance. And it will be extraordinarily difficult to decide whether in single cases one stands before something quite immediate — whether tradition, suggestion and reflection are not at work. There is a tendency to regard states that are habitual and hallowed by tradition as quite immediately given. In all attempts to distinguish between faith and knowledge such a tendency appears in greater or less degree.

Hume too referred to an immediately given. But for him it was not a whole or a continuum, but a chaotic manifold. But a chaos is just as difficult to establish as an absolute continuum. The presence of a chaotic difference-series can be proved only when searching attempts to find order and connection have ended with a negative result. Hume saw, as Bergson did, that the first foundation of cognition cannot itself be rational. Absolute manifold is just as irrational as absolute continuum. Both discontinuity and continuity set tasks — the former that of finding more connection, the latter that of finding more difference. Every subject-matter presents, both within itself and over against other

[8] I have discussed this question already in *Den menneskelige Tanke*, p. 10–19.

subject-matters, both continuity and discontinuity. If we would speak of an immediately given, both categories must be applied.

The very concept "immediately given" is formed by reflection. If there were nothing present but an absolute "immediately given", no concept whatever, and so not the concept "immediately given" either, could be formed. It is formed precisely in opposition to reflection — and by reflection itself, when this becomes conscious of its manner of working and of its presuppositions. Every immediately given is determined by its opposition to a certain degree and kind of reflection. What is immediate in one respect need not be so in others.

It is the need to determine the relations between different subject-matters, and between the elements of the same subject-matter, that leads out beyond the immediately given. In an immediately given (an intuiting) there is perhaps for the time being an equilibrium between gathering-together (synthesis) and relation, between continuity and discontinuity, between likeness and difference. But the equilibrium can be abolished, and there will then be an oscillation between opposed tendencies. The preponderance of synthesis, continuity and likeness leads to continuing in the involuntarily entered track, to acting on the principle "Why not?" This tendency can lead to combining, connecting and identifying on an ever greater scale. But when relation, discontinuity and difference make themselves strongly felt, closer determinations and boundaries are sought, and one acts on the principle "Why?" There is then required a statement of the relations in which subject-matters and elements stand to one another, and an exhibition of how there can be connection in spite of the discontinuity, and of how the degrees of likeness and difference stand to one another. Determinate problems are now set — now, when the intuition's equilibrium has been abolished.

But the different categories that in the formation of problems come to stand in opposition to one another are nevertheless connected in pairs as correlative concepts, and the one member of such a pair cannot find application without the other. A determinate formation of the one member must bring with it a determinate formation of the other. Wundt has in an interesting treatise discussed the significance of correlative concepts and exhibited the tendency, which makes itself felt especially in speculative systems, to keep exclusively to a single one of the members without regard to the other, or perhaps even in such a way that the other's validity is wholly denied. Hegel's dialectic arose precisely from his becoming aware of the mutual connection of the fundamental concepts, according to which the one must with necessity be supplemented by the other.[9]

A pseudo-problem has often presented itself here through the developed consciousness's distinction between subject and object being read into the primitive consciousness. Such a distinction becomes possible only through doubt, which again presupposes disappointment. The scholastic distinction between act of consciousness and content of consciousness has even been read into immediate consciousness, and belief in reality is then explained by the content's gaining influence on the act. But a distinction can be made between act and content only after it has shown itself that there can be "acts" without real content. From the first no difference at all makes itself felt between one's own activity and the content received. This difference arises when one has acted on — that is, drawn all the theoretical and practical consequences of — one's content of consciousness, and this acting then brings disappointment with it. It is the need to act that both grounds the original trust in any given content of consciousness and can later lead to doubting and correcting it.

Of special interest, in the epistemological respect, is that the criterion which can put an end to doubt lies in the question whether the content can constitute part of as great a connection as we are able to establish between our observations and our memories. To take a simple example. When I cannot remember whether I have shut my street door — that is, when I cannot recognize the sensations of touch and movement that the shutting brings with it — it can help if I can recall and recognize all the other circumstances (the grasp after the key in my pocket, the inserting of the key in the lock, the slam of the door — perhaps also what I was thinking of at the moment as I stood at the door, and so forth). It is the whole that abolishes the doubt.

But with an example like this we have already gone beyond the group of the fundamental categories. That shows itself already in the fact that a small and slender, but rational, thread of logical consequence can mean more than a great complex of particulars. In order to understand how this is possible, we must investigate the two following groups of categories, the formal and the real.

[9] Wundt: *Zur Geschichte und Theorie der abstrakten Begriffe* (Kleine Schriften, I, p. 226–258).

Chapter III

The Concept of Totality and the Formal Categories

11. Concept, judgment and inference designate different forms and degrees of logical totality. The concept is a thought-whole in which a group of marks is connected in a determinate way. The judgment is a thought-whole in which a group of concepts is connected according to their mutual relation. The inference is a thought-whole in which several judgments are connected according to the way in which they mutually presuppose one another.

In the inference the formal totality appears most plainly. Every member is bound to the others in the most intimate way, namely through identity as regards a single common concept. And in the relation that thus shows itself to obtain between different judgments, rationality appears. All the premises together contain the conclusion in advance, and each single premise acquires significance only by being connected with other premises. Such an intimate unity of thoughts, in which each single thought bears the others and is borne by them, is philosophy's great ideal. GOETHE's Mephistopheles mocks the way in which formal logic separates out and emphasizes the single members of this whole. But this mockery is warranted only where the "spiritual bond" is lacking. That an organism can be dissected does not prevent it from being a whole so long as it is alive. Nor is it the case that all presentations of formal logic let the thought-whole in its different forms come equally into its own. Language often puts difficulties in the way here, since the very fact that each single mark, each single concept and each single judgment receives its own particular linguistic expression gives them an apparent independence over against one another and an apparent indifference towards the connection in which they occur. It would however be an injustice to language to lay weight only on this side of the matter. For language shows precisely its close connection with thought in this, that the single word and the single sentence become fully intelligible only by being referred to a whole — either the linguistic whole, the entire system of words and sentences to which they belong, or else the whole concrete situation in which they arise.[1] It is always due to an abstraction, often an unwarrantable abstraction, when the single word or the single sentence is torn out of the linguistic or real connection in which it arises.

It is science's task to lead all subject-matters that appear into a great connection, which has its model in the thought-whole our concepts, judgments and inferences constitute. Critical philosophy maintains here, in opposition to dogmatism in all its different forms, that only an advancing carrying-through of a great analogy can be attained, without our having the right to make this analogy into identity. There present themselves nevertheless tasks enough within this way of looking at things. For there can be several different thought-wholes, each holding for its own domain of experience, and the question becomes whether these different thought-wholes can be worked together so as to constitute a single thought-whole. And within every single thought-whole the question arises whether it exhausts the domain of experience to which it corresponds — whether the concept contains everything that "lies in" the intuition — whether the judgment's predicate exhausts the subject's content — whether the conclusion gives everything contained in the premises.

The condition for a formation of concepts is that the subject-matters are ordered in a determinate way, so that their mutual relation can appear. If the concept horse is to be formed, we must have the different races and variations, forms now living and forms extinct, ordered in a series, and the concept then expresses the relation between these different forms, their likenesses and their differences. The formation of judgment again presupposes an ordering of concepts, especially in a partial identity-series, and the inference presupposes an ordering of concepts in one of the rational series.

All such series-formations aim at ordering the content of a provisionally formed group of subject-matters (a subject-matter totality), so that a closer analysis can take

[1] *Den menneskelige Tanke*, p. 84. — *Det logiske Prædikat* (Vidensk. Selsk. Overs. 1914), p. 242.

place. In every single case more will be given than the task set requires. The whole series $A<B<C<D$ I have no use for, if it is merely the relation between B and D that I wish to determine; I then leave A out of consideration, and C I use only as an intermediate member that can be eliminated. The analytic method starts out from the thought of a task, and it is then a matter of finding out what in the series presented can be used and what is superfluous. A discovery is often made by distinguishing between the subject-matters presented and then choosing what is to be used. As Poincaré expresses it: Inventer, c'est discerner, c'est choisir.[2]

12. Formal logic has almost exclusively taken account of concepts that are drawn from intuition and that can be ordered in a graded series, such that the following concept always has less content but greater extension than the preceding. It is classification that has governed it since Aristotle's time. A presupposition here is that the first concepts in the ascending series, namely the individual concepts, are fully finished before one passes over to the general concepts.

But when one considers more closely the relation between the individual concept and the immediately given, the subject-matters it is to correspond to, one sees the necessity of distinguishing between two sorts of individual concept, which in my *Psykologi* I have proposed to call concrete and typical individual concepts. When a subject-matter (or a group of subject-matters) undergoes changes and runs through different stages, one can seek to determine the inner relation between these different stages, the law according to which the one succeeds the other — that is, the law of development of the given subject-matter, of the whole presented. Of each single stage one can have a concrete individual concept; of the whole, as it appears represented by the series of stages, one can have a typical individual concept. When a typical individual concept has been formed, there is present a series of concrete individual concepts in a determinate mutual relation as members of it. So in the concept a biography is able to give us of a human personality: the different stages of development it has run through contribute, in a determinate mutual relation, to the characterization of the personality as a whole. The richer and at the same time the more determinate the typical individual concept is, the better will one be able, going back from it, to understand the single stages in the subject-matter's development. What is significant about the typical individual concepts is that they stand

[2] *Science et Méthode*, p. 48.

to the corresponding concrete individual concepts not as species to individuals, but as whole to parts. The whole contains here the law in virtue of which the one stage succeeds the other. And here content and extension can grow together, which was not possible with the concepts formed in the service of classification.

In more recent times a tendency can be traced to put typical individual concepts in the place of general concepts. To general concepts one can climb up, but from them one cannot, without the help of observation, climb down to the subordinate concepts. Analysis does not lead us from part to whole, but only from whole to part, and general concepts give only a part of the content of the individual concepts. The tendency named appears already in Bruno and Bacon, in that they, as already mentioned, interpret Plato's "Ideas" as concepts of laws. Spinoza rejects the general concepts (*abstracta et universalia*) and demands in their place laws that display themselves in equal degree in the whole and in its parts; in these laws he finds the expression of existence's peculiarity, of something that is fixed and eternal and yet individual.[3] With this it hangs together that he apprehends the whole of nature as one great individual. His concept of nature is a typical individual concept. Leibniz has directly set up the typical individual concept, in that he finds the essence of individuality expressed in the law for the series of its operations (lex continuationis seriei suarum operationum, Lettre à Arnauld 1690; — cette loi de l'ordre qui fait l'individualité de chaque substance particulière, Lettre à Basnage 1698). Later Lotze saw that where a concept expresses the law of a development (ein Bildungsgesetz), the rule about the inverse relation between content and extension does not hold.[4]

The question now is how far such typical individual concepts stand at our disposal. In all domains of experience we must first laboriously form them on the basis of observations that can never be proved to be exhaustive. The characterization of a personality, a people or a period to which historical research can lead, and which can then serve to elucidate the single traits of character and the single events, must itself be won on the basis of report and observation and always stands as a hypothesis. Ethics works by analysis of a typical individual concept, a model (an ideal society, or an ideal personality), which is determined by the fundamental value that bears the ethical train of thought, and by

[3] Hæc fixa et æterna, qvamvis sint singularia, tamen ob eorum ubique præsentiam ac latissimam potentiam, erunt nobis tanqvam universalia. *De emend. intell.*

[4] *Drei Bücher der Logik*, p. 50.

the conditions for asserting and carrying it through. Ethical judgments declare what properties and actions are, in accordance with this model determined by the fundamental value, required of the empirical society and the empirical personality.[5] Epistemology builds on a typical individual concept of human thinking, built on the way this displays itself in involuntary thought-life and in the history of science, and the single categories and fundamental propositions are found by analysis of the concept so formed.

But should there not be a possibility of a thinking in which the typical individual concepts that form the basis of analysis are constructed by thought itself, without the help of observation being necessary at every single step? The whole that conditions the understanding would then be thought's own work. — This question leads us to discuss the relation between logic and mathematics.

13. KANT has already drawn attention to the fact that space and time are not mere general concepts. The single spaces stand to space as such not merely as an individual stands to a species — as the single horse, for example, stands to the species horse — but also as part to whole. Space is composed of spaces. "One can," says KANT,[6] "represent to oneself only a single space, and when one speaks of many spaces one understands by them only parts of the same space." And as it stands with space, so it stands with time too. In these two concepts we have examples of something that — to use the old expression — holds in like manner for the whole and for the parts.

We also find in KANT information about how these typical individual concepts come to be. It happens through what he calls "Synthesis des Gleichartigen". I can go on adding point to point, line to line, surface to surface, volume to volume, according to the same law I began by applying. And number is nothing else than the unity, produced by synthesis, of a manifold given in a uniform intuition.[7]

Space, time and number are accordingly typical individual concepts that we can construct by gathering together a series of entirely uniform differences. It is the unity of the law within the series that makes the construction of such wholes possible.

In analogy with the typical individual representations that lie at the base of mathematics, KANT further, when he sought the presuppositions of natural science, constructed

[5] Cf. *Det psykologiske Grundlag for logiske Domme* (Vid. Selsk. Skr., 6. Række, hist. og filos. Afd. IV, 6), p. 385. — In Chap. V we shall come back to this point.

[6] *Kritik der reinen Vernunft*[1], p. 25.

[7] *Kritik der reinen Vernunft*[1], p. 242, cfr. p. 142 f.

a concept of experience or nature as a whole of observations connected according to law,[8] and it is only on the basis of this construction that he can set up a whole system of fundamental propositions as valid in advance for all phenomena. This appears especially plainly in his grounding of the causal proposition. Experience in KANT's sense is a typical individual concept. Every single special observation leads only to a concrete individual concept. Within experience (or "nature"), in this sense of the word, all subject-matters stand to one another as ground to consequent or conversely. This fundamental logical relation is repeated everywhere, just as within the world of space, time and number it is the same intuitable relation that is repeated at every point. All strict science accordingly builds on typical individual concepts, whose construction (which does not always happen with full consciousness) is possible because it stands in thought's power to repeat itself by always following the same rule of advance.

All these constructions build on a principle by which mathematical thinking separates itself from formally logical thinking. Purely logically, repetition has no significance. It leads to nothing to combine a concept with itself; neither content nor extension is altered thereby. Over against an absolute identity-series pure logic stands just as powerless as over against a chaotic difference-series. But epistemology has more tasks than those set by formal logic; it has also to investigate the presupposition of mathematical and exact scientific cognition. And it is here that the possibility of a positive significance of repetition shows itself. By repeating one and the same act of thought — something we can do without the help of observations — we can produce series of uniform members, which can then be gathered into wholes of different degrees. That was precisely what KANT meant by "Synthesis des Gleichartigen". Whereas pure logic comes to a halt at a chaotic difference-series, mathematics takes its beginning here by drawing attention to the fact that even if all the members are equally different from one another, so that no ordering is possible, the members have nevertheless this in common, that they are all due to repeated acts of thought, and that the subject-matters of these acts of thought are gathered into a whole. And whereas formal logic comes to a halt at an absolute identity-series, mathematical thinking draws attention to the fact that it is nevertheless a series of different acts of thought that has produced the identical concepts. From both sides the possibility of an identically varying difference-series is now seen. Such a one

[8] *Kritik der reinen Vernunft*[1], p. 110; 156, cfr. 2. Aufl. p. 185; 263.

lies plainly at the base of the concept of number, but no less of the series of places in space, of moments in time, and of steps on a scale of degrees, when one leaves out of consideration the qualitative differences between places, between moments and between degrees.

In *Den menneskelige Tanke* (p. 181) I did indeed emphasize the significance of its being possible, by the concepts time, number, degree and place, to eliminate the differences of quality and thereby to prepare the way for a rational treatment which by the road of analogy can also benefit cognition in face of other qualitative differences. But I had not there clearly seen that it is the same principle, the positive significance of repetition, that makes itself felt in all those four categories. Only in the genesis of number had I emphasized this significance, and it is indeed there that it was historically first pointed out and appears most plainly as well.

Leibniz is the first who saw that it is the positive significance of repetition that forms the boundary between logic and mathematics,[9] only that he did not see that even if what is counted is always the same (as with Tordenskjold's soldiers, or with the buckets of a dredger), there is nevertheless in such cases too a series of acts of thought that can be gathered together. Kant lays, as we have seen, "the synthesis of the uniform" at the base of his epistemology and has constructed his concept of "experience" or "nature" in analogy with the concepts space, time and number. It has been thought that Kant was here influenced by Maupertuis, who in explanation of the possibility of mathematics points to the fact that in experience there prevails a certain possibility of repetition (*réplicabilité*), since certain simple natural things recur without essential difference, and that from such experiences mathematics would then draw its concept of a whole arising by simple repetition of a unit.[10] But Kant lays the weight on thought's own capacity to

[9]Si idem secum ipso sumatur, nihil constituitur novum, seu $A + A = A$. Equidem in numeris 4 + 4 facit 8, seu bini nummi binis additi faciant qvatuor numeros, sed tunc bini additi sunt alii a prioribus; si iidem essent, nihil novi prodiret. *Non inelegans specimen demonstrandi in abstractis* (Opera philosophica, Ed. Erdmann, p. 95).

[10]Maupertuis (Histoire de l'Académie de Berlin, 1756, p. 394) finds repeatability (*réplicabilité*) as an actual property of number and extension, and as the only property by which they distinguish themselves from other sensible properties, but he does not go into the process by which number and extension become possible. Cassirer (*Das Erkenntnisproblem in der Philosophie und Wissenschaft der neueren Zeit*, II, p. 334; 495; 725 f), who has drawn attention to Maupertuis's treatise, thinks that Kant found a point of attachment in it. But Kant apprehends repetition as an activity by which pure concepts are produced. And for the time being Kant applies (as Cassirer himself states) this conception only to the concept of number, and only later (after the discovery of the concept of synthesis) to extension as well, whereas Maupertuis

repeat the same act. And only when KANT had by another road[11] found the concept of synthesis could he see the great epistemological significance of the positive character of repetition, as "Synthesis des Gleichartigen" in virtue of "the thoroughgoing identity of apperception".[12] It is remarkable, and it is at the same time an important starting-point for a criticism of KANT's epistemology, that whereas in the case of the typical individual concepts that form the foundation of pure mathematics he expressly and at length treats the question of the warrant for applying them to given subject-matters not constructed by us, he raises no corresponding question with respect to the concept "experience" or "nature" constructed by himself. This comes, no doubt, from KANT's not having been quite clear himself that what he called "synthetic judgments a priori" was one and the same thing with the judgments due to a "Synthesis des Gleichartigen". And this hangs together again with his unfortunate distinction between "Aesthetic" and "Analytic".

In the nineteenth century BOOLE discussed at length the relation between logic and mathematics from the point of view of the positive significance of repetition. Just as little as KANT did he know LEIBNIZ's treatise on this subject (it was first printed in 1840, and BOOLE did not know ERDMANN's edition). He expresses the difference thus,[13] that purely logically $x = x^2$, but mathematically only for $x = 1$ or $= 0$. And in the most recent time HENRI POINCARÉ has thrown light on this relation. Neither experience nor purely logical analysis can, he maintains, explain mathematical induction, that is, proof *par récurrence*, whose principle is that what holds for n holds also for $n+1$. This principle corresponds, according to POINCARÉ, to what KANT calls an a priori synthetic judgment, and by it an extension at once of our concepts' content and of their extension becomes possible. The principle itself has its ground in the possibility of repeating an act, when this act is possible at all.[14]

It is not, however, all repetitions of an act of thought that are equally fruitful. The act of becoming conscious of oneself, of making one's I (its states, its work, its circumstances) into an object for itself, can formally be repeated indefinitely. The I that becomes conscious

speaks of both at once.

[11] See on this *Kontinuiteten i* KANT's *filosofiske Udviklingsgang* (Vid. Selsk. Skr., 6. Række, hist. og filos. Afd. IV, 1), p. 27 f.; 52–54.

[12] *Kritik der reinen Vernunft*[2], p. 133.

[13] *Investigation of the laws of thought* (1854), p. 48 ff.

[14] *Science et Hypothèse*, p. 23. — Cf. however PAUL TANNERY's remarks in *Bulletin de la Société française de Philosophie*, III, p. 126 f.

of itself can itself become the object of a new act of self-consciousness, and so on. Such a series ($S_1 \{ S_2 \{ S_3 \{ S_4 \ldots$) has already been mentioned above in connection with the possibility of an epistemological testing of epistemology. Here we return to it in order to exhibit its difference from the series by which the concepts time, number, degree and place are constructed. Whereas in those four series what is repeated is always of the same quality, so that the following is always a mere continuation of the preceding, in the objectification-series every new member, every new S, is — as an effect of the objectification precisely — more or less different in kind from the preceding. Becoming conscious of oneself is never without effect. It can mean criticism or understanding, but it can also mean subjective entanglement as opposed to absorption in a cause or an endeavour; it can mean greater riches, but it can also mean poverty. It is the historical character of the life of the spirit that excludes the repetition of the act of objectification from acquiring the significance of scientific construction that the four series spoken of possess. The process of objectification can become of great psychological and historical significance, but it yields no new epistemological foundation, as the philosophy of Romanticism supposed.

14. Formal thinking already has its criterion of truth. In pure logic and mathematics everything is true that contains no contradiction. But this negative criterion presupposes a preceding positive unfolding of intuitions and representations, an involuntary, perhaps unconscious process of formation. Here there works that need and tendency, so deeply grounded in the nature of consciousness, towards supplementation and expansion, towards living through a state wholly and fully and towards continuing it as long as possible. The so-called laws of the association of representations are only particular forms of this tendency, which plays so great a part in Hume's psychology of cognition. Relations of likeness and of difference make themselves felt here simultaneously in a remarkable way. For what releases the expansion spoken of will especially be observations and lived experiences that by their degree of strength or by their effect on feeling set consciousness in strong movement. A twofold need then makes itself felt: to hold the subject-matter fast and dwell on it — but also to get full vent for the released energy by going out beyond the subject-matter. This twofold need is released not merely by continuing the state as long as possible, but also by passing over to more or less kindred states. Here the different degrees of likeness, not least analogy, come to make themselves felt. Imagination (in the narrower sense) is a special form of this kind of psychical process. A halt in the involuntary unfolding comes only when in one way or another two subject-matters, or two members in the involuntarily formed series, are held together without being able to be united, since the one pulls down what the other builds up. Then the involuntarily built whole creaks in its joints.

Although the negative criterion can thus come into sharp opposition to the positive unfolding and whole-formation that its very application presupposes,[15] it is itself nevertheless a testimony to thought's synthetic, whole-forming nature. Consciousness of contradiction presupposes that the members that contradict one another are held together in an attempted whole-formation. If the subject-matters each belonged to its own whole, the contradiction would not come to light, at least not before the two wholes had been analysed and their parts attempted to be combined with one another. Synthesis shows itself precisely and plainly in a self-contradiction of which one becomes conscious, just as all states of dividedness in general (doubt, sorrow, remorse) most plainly prove the synthetic character of consciousness. The negative criterion is accordingly the expression of a peculiarity of thought itself. Our spirit is here already the model for everything that is to be capable of becoming an object for it — or, as KANT has expressed it in an earlier note,[16] "Ich bin selbst das Original aller Objecte".

Formal thinking has, however, a positive criterion too. It does not merely thrust out the contradictory as untenable, but itself produces new connections, new immediate relations, through substitution and elimination. Sometimes substitution precedes elimination, sometimes they coincide. If I have two judgments $A = B$ and $B = C$, I can put the expression for B in the one judgment in place of B in the other judgment, and then get $A = C$, since B can then drop out. If on the other hand I have the series $A = B = C$, I push B out, on the presupposition that the B I come to from A is the same as the one I come to from C. It is one and the same process, expressed in two ways.

The negative criterion gives an answer to the question that lies tacitly in the secure, positive unfolding: Why not? Why not go on as hitherto? The positive criterion gives an answer to the question: Why? Why set up this new connection? — Rationality makes

[15] Positive judgments therefore precede negative ones. *Den menneskelige Tanke*, p. 76 f.

[16] *Lose Blätter aus KANT's Nachlass*, I, p. 19. — I have already adduced this passage in *Kontinuiteten i* KANT's *Udviklingsgang*, p. 15.

itself felt along both roads.[17] Inference can be drawn, a transition formed from ground to consequent, in both ways.

Ground and consequent (for example a series-formation and the new relations that can be derived from it) constitute a logical totality. But a complete logical totality is obtained only when the consequent can not merely be derived from the ground, but when the inference can be reversed, so that there is full (i.e. two-sided) logical equivalence between the members of the logical totality. But where the inference has been reached by elimination (whether this occurs after substitution or not), it cannot be reversed. From the series $A = C$ I cannot derive the series $A = B = C$. So long as the premises are richer in content than the conclusion — that is, so long as thought works on the basis of an immediately given that is a greater whole than itself — so long can the logical equivalence that is the logical ideal not be reached. When one excludes in advance all members in the experiences presented that do not enter as members into the conclusion, one can naturally reach equivalence. Thus it stood clear to the founders of modern natural science that one had to abstract from the sense-qualities if one wanted a rational system that could stand as the intellectual expression of the changes in the world. There remains here, then, the question how the relation stands between the intellectual result thus won and the series of qualities whose peculiar properties have been abstracted from. To this question we shall come back in another connection.

The truth that thought can find is a new whole, which replaces the involuntarily formed whole that was the foundation of the thought-work. And the warrant of this substitution rests on the presupposition that truth must constitute a whole coming more fully into its own than it did in the immediately given whole. This presupposition lies at the base of every question that is raised and every task that is set. Within the whole that truth is presupposed to constitute nothing can be lacking that the connection requires; nothing can stand undetermined; no member can conflict with any other member; — indeed the members or parts must reciprocally determine one another and can be derived from one another in virtue of the law that connects them within the whole.

15. Only by the road of analysis does scientific truth come to be, but analysis itself presupposes, as I have tried to show in the foregoing, an involuntary synthesis. As I have expressed it in my treatise on intuition: we live immediately and involuntarily in

[17] *Den menneskelige Tanke*, p. 204–207.

concrete and practical intuitions, but science's road goes through analytic intuitions, that is, through acts of consciousness that establish identity or difference between subject-matters presented. Every single member in a conducting of proof consists in an analytic intuition.[18] By its help we become conscious of what elements are contained in the subject-matters presented. It is the nerve of every coherent train of thought, and marks every one of the transitions within it. Through a series of analytic intuitions there may then perhaps arise a synthetic intuition, which gives us a gathering survey of the roads of thought and their results. But analysis itself, the determination of identities and of differences in the single trains of thought, constantly presupposes a gathering-together of the members that are seen as identical or as different. There takes place accordingly, during the unfolding of cognition, a constant reciprocal action between synthesis and analysis.

It was therefore an impossible and unnecessary distinction that Kant wished to introduce when he sharply distinguished between synthetic and analytic judgments. Every judgment must be synthetic, for there is connected in it something that has not before been connected for thought, even if it now shows itself to be connected in the subject-matters thought is occupied with. And every judgment is analytic, since its predicate must be contained in the subject-matter discussed, if the judgment is otherwise correct. Kant thought, in the case of analytic judgments, of such judgments in which the subject (the initial representation, the subject-matter) constitutes the whole totality that is analysed. But it can make no difference of principle here whether it is a single subject-matter or a plurality of subject-matters that I analyse. The restriction to a single subject-matter, a single initial representation, will always be more or less artificial. When I say "the man is good", and form this judgment at first hand, I find the occasion for the predicate "good" in an observation of the man's way of acting in different situations, neither in the abstract concept nor in the concrete image of intuition of the man. I bring the man into different relations. And so too, to use Kant's favourite example, when I say "bodies are heavy". I have then seen the single bodies as members in a whole system of bodies, and it is by analysis of the relations within this system that the predicate "heavy" emerges. It is always typical individual representations that are analysed in the so-called

[18] In the treatise named I have mentioned Descartes as the one who has plainly set up the concept of analytic intuition. But it has not disappeared since his time. It is found both in Locke (*Essay* IV, 2, 1) and in Hume (*Treatise* I, 3, 1. 3).

"synthetic" judgments. Therefore, as we have seen above, KANT had to construct the typical individual concept "experience" or "nature" in order to be able to derive his "a priori synthetic" judgments.

The so-called analytic judgments are according to KANT to bring nothing new. But why then do we form such judgments? What need do they satisfy? Judgments are originally predicate-judgments. It is the predicate that brings the new. Therefore language can have sentences in which no logical subject is expressed. — KANT concedes accordingly that even the "analytic" judgments bring something new. For he says in the second edition of the Critique of Reason that in such judgments the predicate is contained in the subject "versteckter Weise" or "verworren". In the first edition the same thing is expressed by saying that in analytic judgments my concept of the subject is "aus einandergesezt und mir selbst verständlich gemacht".[19] But then there is indeed something new here: the hidden is brought out, the confused clarified, and only now do I understand my own concept. It is only a question whether this can be reached without going beyond the single subject-matter and bringing it into connection with other subject-matters. There are really no purely analytic judgments at all. Not even the judgment $A = A$ is purely analytic. This judgment is a definition of what a concept is, and at the same time an assertion of what is the presupposition of every conducting of proof, every analytic intuition. The whole that lies at the base of its being set up is the demand that the conducting of proof makes on the representations that are to enter as members into it. The proposition lies accordingly "versteckter Weise" in every inference, every attempt at proof, not in A in and for itself.[20]

As we have seen above (§ 13), mathematical fundamental propositions are formulated on the basis of an intuition or "induction" that can always be extended in virtue of the same law or tendency according to which it came to be. POINCARÉ has, it seems to me, clearly refuted modern attempts to apprehend the mathematical fundamental propositions as "analytic" in KANT's sense of the word, and he has interpreted KANT's

[19] *Kritik der reinen Vernunft*, 2. Udg. Einleitung § 4. — First edition: Einleitung. Von dem Unterschiede analytischer und synthetischer Urteile.

[20] In more recent times the distinction between synthetic and analytic judgments has repeatedly been discussed in the French philosophical society. Bulletin II (1902): COUTURAT: *Sur le rapport de la Logique et de la Métaphysique de* LEIBNIZ. Bulletin III (1903): PAUL TANNERY: *La valeur de la classification kantienne des jugements en analytiques et synthétiques.* COUTURAT would apprehend all judgments as analytic; P. TANNERY maintains that every judgment can, according to the point of view, be regarded as analytic or as synthetic.

"synthetic propositions a priori" as grounded in analysis of "mathematical induction" (*loi de récurrence*).

How far this analysis is to go, when it is a question of setting up the first definitions in geometry, is a question that has been answered differently. On the strict Platonic-Euclidean conception one must here go to the limit — to the point as the limit of the line, to the line as the limit of the surface, to the surface as the limit of the body — and the mathematical propositions hold only for these ideal subject-matters. The tangent touches the circle in only a single point, and between two points only a single straight line can be drawn. In antiquity Protagoras[21] objected against such a conception that every actual tangent touches the circle in more than one point, and referred throughout to the sensible lines instead of to the ideal ones. In the most recent time Hjelmslev has given a presentation of elementary geometry that does not set up the ideal definitions, but starts out from the lines and figures that can be drawn; and in these a tangent always has a whole stretch in common with the circle, and there can be more than one straight line between two points. He contents himself with saying that there is *at least* one straight line passing through two given points, and that the foot-point of the perpendicular on the tangent must be contained in the stretch that circle and tangent have in common.[22] Whereas the Platonic-Euclidean conception goes to the limit in its determination of the concepts point and line — a limit that is never reached in our intuiting — a conception like Hjelmslev's holds to the fact that we actually move in the domain of intuition, and that with all the tangents we have to do with there will always be several points common to the tangent and the circle. The main thing is that the foot-point is contained in the stretch common to circle and tangent.[23] — In the last resort the difference between the two conceptions is not great, but it is of epistemological interest, since it throws light on thought-work's relation to intuition. Both conceptions hold fast that we do not in fact get beyond intuition; but the one lays weight on the approximation to the ideal standard, while the other lays weight on the ideal having its place in the intuited itself, although intuition contains more than the ideal.[24] Whether Protagoras thought of the matter in

[21] Diels: *Fragmente der Vorsokratiker*[1], p. 520.

[22] Hjelmslev: *Om den rette Linies Bestemmelse ved to Punkter* (Vid. Selsk. Overs. 1916), p. 183.

[23] Hjelmslev: *Elementær Geometri*, I, p. 31.

[24] Perhaps Hobbes' definition of a point might be of interest in this connection. *De Corpore* XV, 2: — per punctum non intelligi id, qvod qvantitatem nullam habet… sed id, cujus qvantitas non consideratur, hoc est, cujus neque qvantitas neque pars ulla inter demonstrandum computatur.

this way, or merely used his conception for sceptical ends, must be left open.

Chapter IV

The Concept of Totality and the Real Categories

16. The wholes that are formed by help of the formal categories are at once the expression of the nature of human thought and the models for the activity by which subject-matters that thought has not itself produced are worked up. Thought cannot, in its working activity towards the subject-matters, deny itself. Therefore the real categories presuppose the formal, just as these presuppose the fundamental. This follows plainly from the fact that the formal (logical-mathematical) fundamental concepts contain the foundation of all understanding, whatever subject-matters it may concern. To understand something in the stricter sense is not merely to recognize it, but to see it as a consequence of something already known. A relation between ground and consequent is accordingly presupposed. This relation can again be of such a kind that there is only one-sided equivalence, that is, that the relation cannot be reversed, or that two-sided equivalence holds, so that one can make the consequent into the ground and the ground into the consequent. Therefore we see also that in the case of the most important of the real categories, the one we keep to for the present, causality, it is a matter of exhibiting one-sided or two-sided equivalence between two subject-matters. The presupposition is that of the subject-matters in question rational series can be formed.

Where such series cannot be formed, and therefore no relation of equivalence, neither one-sided nor two-sided, can be formed, strict science is not possible. There can only be a question of ordering, division and comparison. The ground why rational series cannot be formed lies in the last resort in the changing qualities of the subject-matters. It is not the qualities in and for themselves, nor the time-relation in and for itself, that makes difficulties.[1]

The qualities might be thought of as existing once for all, realized in all possible extents and in all possible degrees, forming a great, reposing system. In SPINOZA's system this possibility is seized, as Greek thinking had moreover aimed at all the way from PARMENIDES. SPINOZA first reduces all differences of quality to the one difference between spiritual and material, and these two qualities are apprehended as coordinate attributes of existence. In such a system there can only be a question of formal wholes. All relations within each single attribute (SPINOZA assumes there are infinitely many besides the two we know) are logical-mathematical. The relation between ground and consequent becomes the only one holding — and the consequences need not even be drawn; they are eternally contained in their grounds, in the last resort in a single ground.

The time-relation in and for itself could also be thought of as purely rational. According to the concept of time in its purity, every moment lies between two other moments, and since the transition between any two moments whatever takes place in quite the same way as between any two others, we get an identically varying difference-series, within which the single members present no difference other than that consisting in their different place in the series. Time-totalities (groups of moments) can be ordered in such a way that the one stretch of time not merely always lies between two other stretches of time, but also in such a way that a progressive difference-series is formed, in which every member is greater (or smaller) than the preceding and smaller (or greater) than the following. The pure concept of time accordingly permits rationality very well.[2]

Neither differences of quality by themselves nor differences of time by themselves conflict with rationality. But it is the changing of the content of time, the alterations of quality, that brings the problem with it. As regards content, one cannot without more ado infer from the one moment or the one stretch of time to the other. Is it possible, on

[1] What follows is a correction of the presentation in "Den menneskelige Tanke", p. 174 f.

[2] KANT does not let the pure concept of time come into its own as the foundation of a little rational world of its own, which in its formation nevertheless needs analogy with space and the application of the concept of magnitude. He parallels it too much with space and is inclined to put it in close relation to number. In *Kritik der reinen Vernunft*[1], p. 142 f., number is defined as the unity in the gathering-together of a uniform manifold that is conditioned by my producing time by analysis of intuition. In *Prolegomena* § 10 time is laid at the base of arithmetic, as space is of geometry.

the basis of alterations of quality as given, to build strict inferences (with one-sided or two-sided equivalence)?

If we look at the history of science, it shows itself that thinking has taken the road of working through certain important, constantly recurring differences of quality in such a way that they can yield schemata for the changing qualities, and in such a way that series can be formed within which every member is determined by its place in relation to the other members. Such schematisms have been formed, besides by the concept of time, by the concepts number, degree and place.[3] It is to them that exact empirical science owes its coming to be. The differences of quality (both the simultaneous and the successive) are ordered in series of such a kind that the members stand to one another as the members in one or more of those schemata.

But the relation between schema (for example pure time or pure number) and subject-matters (for example qualitatively filled time, or qualitatively different phenomena) is not identity, but analogy. For observation, every moment is and remains filled in a way that is qualitatively different from the way in which other moments are filled; and whereas transition from one number to another can always take place in the same way, the transition from one quality to another is and remains different from the transition between two other qualities. The content of time can be altered by the transition from one moment to another, although time's law is the same. And whereas, for example, the transition from one degree of refraction of light to another is purely quantitative, and the transition forms a progressive difference-series, the transition within the qualities of the colour-sense forms only a regularly varying difference-series. And yet the analogy can be carried through, since to a certain degree of refraction (a certain quantitative difference in the refraction of light) there always shows itself to correspond a determinate quality of colour (within the spectrum). By the road of analogy the colour-series has become rational.

Kant saw this clearly, as regards a single relation. In comparing the purely logical-mathematical relation between ground and consequent, which is not a time-relation at all but rests throughout on identity, with the actual changes we seek to explain as causes or effects of one another, he found only analogy between them — an analogy which, as we have seen above, he grounded as valid from the concept of the possibility of experience.

[3] *Den menneskelige Tanke*, p. 180–201.

Between formal and real science there is, according to Kant, analogy, not identity. The world of qualities and successions constantly makes itself felt anew, even if science decrees that quality can be reduced to quantity and succession to identity. Between the two series formed by the pure concepts (time, number, degree, place) on the one side and the qualitative changes on the other, only one-sided, not two-sided, equivalence can be exhibited. By regarding the series of qualitative changes as if it presented only transition in time, number, degree and place, one can anticipate and predict future qualitative changes. But one cannot prove that no other presuppositions than those determined by those four formal concepts could lead to such results. The difference between a relation of analogy and a relation of identity is precisely this, that by a well-chosen analogy one domain can serve for the intellectual elucidation of another, but it cannot, as where there is a relation of identity, be proved that no other elucidation was possible.

Causality can only be defined as a succession whose members stand to one another as ground stands to consequent. Out of the two elements in causality, the logical and the temporal, we cannot by combination form a new, positive concept. We do not really know what cause is, if we are not content with the definition given. A causal relation is a totality that subsists through the constant analogy between the logical and the temporal elements — an analogy which, as a working hypothesis, invites us to overcome the difficulties for understanding that the alteration of quality brings with it, by conversion into purely logical and mathematical relations. So long as the need for cognition is awake, work will be done along this road to produce real counterparts to the purely logical and mathematical totalities. But whether this is possible in face of all the subject-matters presented — whether the concepts "experience" or "nature", as Kant had constructed them, became more than ideal hypotheses — that question he did not raise. And only for that reason could he think he had solved Hume's problem.

The general principle that in earlier times was called the principle of sufficient reason, but that now appears to us in its formal form as the principle of grounding and in its real form as the principle of causality, is itself a consequence of the necessity of thinking all subject-matters as members in a whole. A need for the supplementation of the single subject-matter makes itself felt here. Telesio already, later Herbart and Hamilton,[4]

[4] *Den nyere Filosofis Historie*, I, p. 93 f.; II, p. 248, 388.

and in the most recent time Bradley,[5] have clearly seen this.

17. The theory here put forward can be criticized from two different sides. One can deny the warrant and the necessity of reflection's working-up of the subject-matters, and one can maintain that this working-up leads to such a carrying-through of logical and mathematical points of view that there is no ground to speak of mere analogy. In the first case the analogy falls away because one keeps to the subject-matters in their immediate form, without interpretation; in the second it falls away because the subject-matters are wholly converted into members in identical judgments or principles. — As types of the first conception may be named Henri Bergson, of the second Hermann Cohen.

For the first standpoint the train of thought is roughly the following. We must make ourselves familiar with the thought that what is immediately given to us is qualitative change, and we must keep to this immediately given, abstracting from all abstractions and reflections. What matters is to live oneself into reality, and reality is not in our interpretations and thoughts, but in what immediately announces itself to us. The great apparatus that science constructs for the explanation of reality is very serviceable for making things serve our practical ends and for communicating ourselves to one another; but it leads away from what is really given. This apparatus lays claim to us for practical reasons to such a degree that it can be very difficult to push it aside and to free oneself from all the abstractions and reflections that partly lead to it and are partly occasioned by it. But if that succeeds, all problems fall away.[6]

This conception starts out from the assumption that a distinction can be made once for all between subject-matter and reflection. But this difference varies infinitely, both in kind and in degree. And one can trace already in the immediately given the laws and forms according to which reflection works. The stamp of totality is already present in it, and reflection only unfolds further what is already intimated in it. (Cf. above § 10.) Such a further unfolding is necessary on account of the contradictions and oppositions that the immediately given can already contain. There is not, after all, a single immediately

[5] *Essays on Truth and Reality* (1914), p. 312: Where *A* is not real by itself, but implies and belongs to an ideal whole, you want a reason for *A*, for you want to known the How of this unity… But the "axiom of ground" is a result and not a premise… The "axiom" holds only so far as a thing is not complete in itself… The demand for the making good of such imperfection,… the completion of the thing in idea so as to satisfy us theoretically, is what we mean by the search for "why" and "how".

[6] Cf. my book *Henri Bergson's Filosofi*, p. 16 f.; 58–60. — William James stands close to Bergson on this point. See *A pluralistic Universe* (1909), p. 242–247; 260–263.

given, a single concrete or practical intuition, but many different intuitions, each with a character of wholeness — and how can these different intuitions be united into a more comprehensive intuition? Without reflection and comparison it cannot happen.

In my criticism of Bergson in "Henri Bergson's Filosofi" I kept to the relation between intuition and reflection in general. But the relation of the many different intuitions to one another also leads out beyond the immediately given, even if one concedes that we can get back to it.

There presses forward, furthermore, the necessary question with what right one calls the immediately given "reality". The difference between imagination, or dream, and reality does not exist for concrete and practical intuitions. The problem this difference sets exists there just as little as any other problem. A grounded difference between reality and dream presupposes cognition of a fixed, lawlike connection between the observations that stand at one's disposal — even if this cognition is ever so elementary and childlike. And it is precisely the need for marks of reality that leads to the cognition of laws. Henri Poincaré was therefore right in the proposition: "Le monde Bergsonien n'a pas des lois" (Dernières Pensées, p. 31).

In sharp opposition to Bergson's theory of intuition stands the Marburg school's theory that the problem of causality is completely solved if one succeeds in working through the differences the given presents in such a way that all relations within the given can be expressed in identical judgments or equations. The mathematical concept of function then contains the solution of the problem of causality. A mathematical function designates a mutual dependence between two variable magnitudes and thus expresses at once their difference and their connection. When $y = f(x)$, x does not fall away because y enters, but is preserved in y. When a relation between two variable phenomena can be expressed in this way, the externality in the popular conception of the causal relation falls away, and there is no longer any question of an outer intervention or action of one thing on another. The essence and relations of things have found their expression once for all in the equation.[7]

However, Cohen does distinguish (p. 248) between the equation and its interpretation; the equation forms only the methodological foundation. It is interpreted as the

[7] H. Cohen: *Logik der reinen Erkenntniss*, p. 185; 239–247. — To a similar result comes Richard Avenarius from his pragmatic standpoint in his theory of absolute "Deproblematisierung". (See "Moderne Filosofer", p. 103–104.)

expression of constancy, subsistence (Erhaltung). But the equation $y = f(x)$ can mean a great many things, can find application in the most various domains. When it is to be the methodological foundation for “subsistence” or “physical structure”, then we have here precisely what KANT called analogy. There is only analogy between mathematical and physical relations, just as there is only analogy between logical grounding and causal relation. It is on the warrant of this analogy that mathematical physics rests, or rather: the warrant of this analogy is the great hypothesis that all modern natural science seeks to establish in detail.

COHEN acknowledges further (p. 412 ff.) that problems arise only where a new subject-matter comes to be for us. It is then a matter of converting this new qualitative content into quantitative form. Sensation makes a demand, but only thought can decide the demand’s significance. Sensation is not to be heard, but interrogated. But the result of these interrogations cannot be established once for all. Our concepts can therefore never become perfect. Hereby so much is conceded, that the carrying-through of the purely logical conception of reality constantly stands as an ideal or a hypothesis. But even where the interpretation must be said to have succeeded, as when the theory of heat has become a part of the theory of motion, is the difference thereby abolished between the quality of sensation and the mathematical formulae that express heat as a physical subject-matter? Sensation’s crutch, says COHEN (p. 425), is shattered by the theory of heat’s having become a part of the theory of motion. But sensation in its immediate quality needs no crutch at all, and even after the theory of heat has become a part of the theory of motion the question stands unsolved: why and how does the phenomenon that physical heat-theory describes by help of its equations make itself felt for sentient beings in a determinate, qualitative way? Can one derive from the equations of physical heat-theory the qualities of warmth that are immediately given to us? One can declare these qualities subjective, but they are not thereby got out of the world. What holds for the temperature-sense holds also for all other senses. And it is not merely the relation to human senses that sets the problem anew. In spite of the proposition of the subsistence of energy, light, heat, mechanical movement, electricity and so forth are nevertheless different natural forces, and the objective difference between them cannot be derived from the general proposition that to a certain quantum of the one there corresponds a certain quantum of the other. The one has in fact other effects than the other, and in every single case there is a transition in a determinate direction. Even if the proposition $A = B$ holds for a certain point of view, it can nevertheless in the single case be of decisive significance whether it is A or B that is present. The question becomes finally whether reality can be thought to consist in the bare equations. COHEN is not far from this consequence. Only thought, the concept, he says, gives true being (p. 180). All true philosophy begins, according to him, with PARMENIDES. Yet he does not seem willing to draw the consequence that only the world of pure identities exists.

Still less than KANT's doctrine of analogy is COHEN's doctrine of identity able to solve HUME's problem. But COHEN does not acknowledge the warrant of this problem at all, laying weight as it does on the differences of quality and of time. He dismisses (p. 230) HUME's whole standpoint as immoral! It must then also have been immoral of KANT to let himself be woken from dogmatic slumber by HUME's problem. And it is consistent, then, that one calmly lies down to slumber again!

18. The concept of cause appears, on the conception I have here maintained, as an extension of the thought-whole that can be reached by the logical-mathematical road — an extension that takes place by qualitative and successive differences being brought, by the road of analogy, under logical and mathematical points of view. There is then formed a real thought-whole, which gives the most perfect cognition of reality that can be won, without reality being identified, in the Eleatic-Platonic manner, with logic and mathematics. That it is only by the road of thinking that we can ground our belief that we stand before a reality cannot be interpreted to mean that this reality consists only in the forms of thinking itself. The relation between observation and interpretation never becomes identity. HUME's problem remains.[8]

Cause and effect, which stand for the time being as two different things, come into a more intimate relation to one another as members in a real thought-whole formed with the formal thought-whole as its model, and in virtue of the tendency to wholeness that

[8] In ANTON THOMSEN's book on HUME an attempt was made to show that the credit for having set up the problem of cause belongs not to HUME but to HOBBES. It is however only a single passage in HOBBES that can be appealed to here (De Corpore I, 3, 20). It was meritorious to bring this passage forward, but what HOBBES has said here gains no influence on his dogmatic train of thought. And before HOBBES too there are intimations of the difference between ground and cause. So in BRUNO (De la causa, principio, et uno. Secondo dialogo: the difference between principio and causa. ED. LAGARDE, p. 230), and still more plainly in the Nominalists of the fourteenth century, especially in NICOLAS D'AUTRECOURT. (See LAPPE: Die Philosophie des Nicolaus von Autrecourt. Bonn 1905. — H. RASHDALL: Nicholas de Ultricuria, a medieval Hume. Aristotelian Society. Proceedings 1906.)

expresses itself in all conscious life, and especially in all thought-life. But from other sides too the concept of the whole, or the tendency to wholeness, displays itself in the causal relation, and it may serve to elucidate our subject to bring these forward.

a. The whole that cause and effect constitute is itself a member in a greater whole. Every single, isolated causal relation (if such a thing exists) is blind and accidental; the condition for two events at this determinate place and at this determinate time being able to stand in a causal relation must be sought in a greater connection, in the last resort in the lawfulness that lies at the base of the whole natural connection we know. In seeing two events in causal connection one sees them at the same time as comprised in a whole more comprehensive than the one they together constitute. This comes out especially when one lays weight on the direction of the causal relation, a side of it that one is inclined to overlook. Leibniz definitely drew attention to direction and set up the proposition of the subsistence of direction.[9] Later the concept of direction has receded into the background, although Kant laid very great weight on the causal relation's differing from a plain and simple relation of succession in that the members cannot be exchanged. Where there is two-sided equivalence between cause and effect, one can go both forwards and backwards, experimentally and in thought. But in actual observation, in the single determinate case, one goes either forwards or backwards; there the equivalence is always one-sided, and it is of decisive significance which of the members is cause and which is effect. The discovery of the equivalence of natural forces has by no means, as Avenarius, Riehl and Cohen think, solved Hume's problem.[10] — In the long run the equivalence is perhaps actually one-sided after all; that at least is the conclusion some draw from Carnot's law.

If it is only against the background of a more comprehensive whole that direction can be established, and thereby the determinate, concrete causal relation seen in its peculiarity — and if the criterion of reality consists in the causal connection, the determinate order and direction in which the single subject-matters arise — then "reality", real truth, becomes an ideal that at any given time can be reached only approximately. This does not however lead, as has been thought,[11] to complete scepticism; it is only a testimony — the most

[9] *Opera philosophica*, ed. Erdmann, p. 132 f.

[10] Cf. *Den menneskelige Tanke*, p. 282–287.

[11] Cf. Bertrand Russell's criticism of the dynamic concept of truth in his treatise *On the Nature of Truth* (Proceedings of the Aristotelian Society, 1907, p. 28–49). Russell himself holds an "analytic realism" that

decisive of all testimonies — that truth develops historically.

b. Just as the ground is always a sum-total of presuppositions, so the cause is always a sum-total of positive and negative conditions. Since now every one of these constituents of "the cause" has a tendency to work in a determinate direction, and since the relation between these tendencies can be highly various, there arises the possibility of very different orderings and combinations within what is designated as "the cause". The cause as a whole can therefore appear in highly various ways, and it will be difficult, if not impossible, to formulate a law that exhausts the actual relations. "A law of nature can", says Ernst Mach, "contain no more than a comprehensive and condensed report of facts. Indeed, it contains even always less than the fact itself, because it does not reproduce the whole fact, but only the side of it important for us, while completeness is, by intention or of necessity, abstracted from."[12] Bergson goes so far as to call the causal laws mere labels. This disproportion between causal law and actual happening appears not least in the spiritual domain. An activity of will whose end is embraced with full consciousness does not unfold without a quantity of involuntary tendencies of a spontaneous, reflex and instinctive kind working with it, besides that representations and feelings not directly concerning the end make themselves felt. And the relation between the conscious end and the half or wholly unconscious tendencies can in single cases vary to a high degree, so that it is no wonder if the causal explanation of a human action is often so extraordinarily difficult.[13]

c. What thus holds within the single "cause" appears in an analogous way when we consider different causal series in their mutual connection. No causal series can be exhibited that runs its course quite in isolation. Different causal series interlock with one another, and what is presented in single cases will always be a complex of causal series whose directions meet in a harmonious or a disharmonious way. —

There presents itself accordingly, unceasingly, the task for research *a)* of finding the great connection within which the single causal relation can find its grounding, — *b)* of determining the relation between the elements within the same member in the single

assumes a world of absolute existences, partly universal (on the model of Plato's Ideas), partly sensible (*Le Réalisme analytique*, Bulletin de la Société Française de Philosophie, 1911). Only thereby does he think he has firm ground under his feet.

[12] *Die oekonomische Natur der physikalischen Forschung* (Populär-wissenschaftliche Vorlesungen[4]), p. 224 f.

[13] Cf. my treatise *Begrebet Villie* (Mindre Arbejder, Tredie Række), p. 40 f.

causal series, — and *c)* of unravelling the relation between the different causal series that together determine a result.

Real wholeness is therefore (as indeed all the categories are) an “Idea” in the Kantian sense of the word, that is to say a thought that constantly sets new tasks and cannot be realized in any intuiting or in any completed train of thought. It is a working thought, which solves the tasks presented only in such a way that it at the same time draws attention to new tasks.

19. If it is right, as has been sought to be established in the foregoing, that human thought in virtue of its own nature seeks to form wholes, because it itself works involuntarily according to a law of wholeness, then it would be a happy thing if there were subject-matters that from the very first, prior to all elaboration by thought, appeared with the stamp of wholeness, and in which it was a law of wholeness that determined the reciprocal action of the single elements. It would then not be thought-work that produced the whole, but the subject-matter itself that came forward as a given whole, having the law written in itself and not needing to fetch it from thought. Now, as we have seen above, all subject-matters are given as wholes; but the thought-work we have hitherto investigated aimed essentially at resolving the given into its elements and winning understanding through a thought-whole brought about by analogy with the logical-mathematical totality that thought can produce on its own. Such subject-matters occur in the organic, the psychical and the social domain, and the question then becomes what research’s relation to them is. Are wholly new categories necessary here, and if so, in what relation do they stand to the categories spoken of in the foregoing?

When totality is set up, as it has been in “Den menneskelige Tanke” (p. 218–227), as a special category, as a single member in the series of categories, that is nevertheless not merely on account of the position of biology, psychology and sociology within science, but also on account of the connection between the concept of cause and the concept of the whole exhibited above — a connection that does not appear only in the three domains named. Everywhere research advances, it discovers wholes that had hitherto concealed themselves. Otherwise thought’s own character of wholeness, its peculiarity as synthesis, would indeed be a highly unserviceable organ. But it has shown itself to be a pure abstraction when one apprehends cause as something single, cause and effect as two different things, and the causal series as lying quite outside one another. When the concept of the whole is especially characteristic of the three sciences named, it does not for all that come forward there as a *deus ex machina*.

It does however make a considerable difference whether the whole shows itself only as a goal that is reached when research can find connection between different causal series, or appears as a subject-matter presented, which then sets research the task of finding its inner and outer conditions. In the latter case it shows itself in a quite special sense, what holds more or less in face of every subject-matter, that research proceeds analytically. Even if a subject-matter appears in ever so high a degree as a whole, understanding is nevertheless not thereby given. There must be analysis, distinction between parts and properties, if the law of their connection is to be found. The wholes actually presented are so many tasks, first for description and classification, then for investigation with respect to the conditions of their subsistence and development. The concept of the whole can never be used for explanation. It is, like every category, both a form of thought's work and an ideal sought to be carried out in detail. But with immediately given wholes there is, as experience right up to the most recent time shows, a temptation to find the explanation in the concept of the whole itself, to use it as a key that unlocks everywhere. All theological, vitalistic and spiritualistic explanations rest on such a misuse. In the concept of God, as it is developed in the higher popular religions, all causality in the world is concentrated as in an infinite unity or whole, and it then stands as a matter of course that every single phenomenon has its explanation by it. Galilei already drew attention to the fact that precisely because everything has its ground in God, it brings no understanding to appeal to God as the cause of a single phenomenon in its peculiarity at a given time and place. Vitalism regards the organism as a being that has its cause in itself, and this is the real cause of everything that goes on within it. In the psychological domain there corresponds to this the making of the soul the cause of its own states, instead of finding the laws of mental life through the study of the relation between the different mental expressions. In the sociological domain a similar tendency appears when one makes society as a whole the cause of special social phenomena.

In Leibniz' concept of individuality there lies a direction for at once acknowledging the peculiarity of a given whole and finding its essence in the law that empirically expresses the connection between its different states and changes.[14] As we have seen

[14]Cf. *Den menneskelige Tanke*, p. 131–135.

above, cognition's ideal was for Leibniz to lead all judgments back to identities, and it might seem to conflict with this that he lays so great weight on individual wholes and their differences. But the point is that identity, which for Plato stood as an object of contemplation — a contemplation approaching mysticism — was for Leibniz a form of thought and a means of thought, used at every single little step, every little transition thought makes in the analysis that leads to finding the law of the individual whole (cette loi de l'ordre qui fait l'individualité de chaque substance particulière; — lex continuationis operationum). This thought in Leibniz hangs closely together with the fundamental thought of his infinitesimal calculus, since it is through infinitely small transitions between the states that a being's continuity expresses itself, and since a being's peculiarity consists precisely in the law according to which these transitions take place.

Leibniz's thought can guide us in our consideration of the organism, the personality and society from an epistemological point of view.

20. In the biological domain mechanicism and vitalism stand over against one another.

The former sees in the organic phenomena merely a series of physical and chemical processes whose laws have at bottom been found, so that it is merely a matter of applying them to special cases. The organism is a sum of mechanically (physically-chemically) working elements. Scientific biology begins with Lavoisier's and Laplace's discovery (1780) that animal heat can be explained by the physical-chemical road. Since then such an explanation has been carried through step by step for various functions, among others for reproduction. There are still gaps left, such as the chemical constitution of the enzymes and of the substances that condition the heredity of a property. But a peculiar principle of life is superfluous, and an organism can be defined as a chemically working machine possessing the peculiarity of being able to develop, maintain and reproduce itself automatically. That a part is so adapted that it serves the whole is for the mechanistic conception only an obscure expression of the fact that a species can live only when it possesses the mechanism that self-maintenance and reproduction presuppose. Extraordinarily many forms have perished for want of such a mechanism. There is nothing to indicate that the artificial production of living matter lies beyond the possibilities of science.[15]

[15] Jacques Loeb: *The recent development of Biology* (Congress of Arts and Sciences, St. Louis 1904, Vol. V).

Even if ever so much weight is laid on the strict carrying-through of physical and chemical methods within biology, it must nevertheless, as even Loeb concedes, be maintained for the present that not everything in the organic domain is explained along this road. At this point therefore the opposing conception, vitalism, quite naturally takes hold. By vitalism one must, if one wants a clear concept and at the same time takes account of the history of biology, understand the conception that not merely denies that organic life can be explained by the physical and chemical road, but also finds the explanation in a special vital force different from all other natural forces.[16]

Vitalism is on this definition both a negative and a positive concept. It rests first and foremost on the presupposition that a mechanical origin of life is not possible. But is it possible to conduct a proof of this? When biology's advances, as regards causal explanation, have always consisted in determinate physical and chemical causes being exhibited for organic processes, it will lie nearest to presume that this advance can be continued as long as research goes on at all, even if a conclusion should not be reachable here. We stand here perhaps at an irrational relation, as with the relation between the quality of sensation and the equations of mathematical physics. (Cf. § 11 and § 15.) All science really works at the analysis of given wholes. A gap in the course of the physical and chemical processes within the organism designates only a new task, the necessity of more decimals. Thus Hugo De Vries too holds fast that the mutation theory, which assumes a sudden coming to be of new organic types, sets precisely the task of finding out the chemical cause of this deviation from the type hitherto prevailing.

— *The Mechanistic Conception of Life*, Chicago 1912. — Loeb acknowledges the significance of Arrhenius' thought that very small germs are driven through space by radiation pressure, and that when they meet bodies of suitable temperature and provided with water, salts and oxygen, they call forth new organic developments. But he thinks that science must nevertheless work to produce living matter, or at any rate to prove why it is impossible.

[16] See my treatise *Om Vitalisme* (Mindre Arbejder, Første Række), p. 44–46. The classical vitalists are Van Helmont and Stahl in the seventeenth century, and the Montpellier school, especially Barthez, in the eighteenth. (Verworn is wrongly named as a vitalist on p. 46.) — I do not understand how Driesch (*History and Theory of Vitalism*, 1914, p. 59 f.) can regard Blumenbach as the most characteristic representative of the old vitalism. Blumenbach regards, as Driesch himself emphasizes, the formative tendency (*nisus formativus*) not as a cause but as a result whose cause is unknown. Blumenbach (*Über den Bildungstrieb*, 1789, p. 25 f.) says expressly that the word "Bildungstrieb" contains an explanation just as little as the word "attraction" does in Newton, and he recalls Newton's express declaration in that respect. These words designate the phenomenon that is to be explained, not the cause. When Driesch therefore says that if all vitalists had expressed themselves as clearly as Blumenbach, Lotze's and Claude Bernard's criticism would have been unnecessary, this is quite right; for there would then have been no vitalism at all!

Positive proofs of vitalism Driesch, the investigator of our day who has most fully maintained vitalism, thinks he has found, partly in embryology, partly in instinct. Morphology is vitalism's real domain. The organic forms with their character of wholeness seem incapable of being derived as the result of a mere interplay of inorganic elements. Driesch refers, as regards embryology, to the fact that in the earliest stages of embryonic development the cells' mutual ordering can be completely altered by mechanical intervention, and yet the cell-complex can, in spite of the splitting, develop in a quite normal way. But, as Haldane remarks,[17] there can from a physiological point of view be no doubt that special physical and chemical causes are at work here, even if they have not yet been exhibited. And in any case the apparently positive proof is only a special form of the negative one, which has shown itself impossible to conduct. — As in the formation of form, so now, according to Driesch, the organism is to work as a whole in the instinctive actions, to lead and determine the direction and connection of the processes, now by inhibiting a single process, now by altering the direction, now by rearranging the relation between the single processes. Such an inhibiting, altering and ordering intervention does not, according to Driesch (who here uses a thought already expressed by Descartes in favour of the soul's intervention in matter), conflict with the proposition of the subsistence of energy. But neither do we get here beyond a negative assertion. An alteration of direction without material cause would certainly not conflict with the subsistence of energy, but it would conflict with the strictly physical conception of the law of inertia.

Driesch's treatment of the relation between the mechanical and the organic is in this connection of special interest, because as a logician he lays strong weight on the category of the whole.[18] But it has in the foregoing, it is hoped, been sufficiently exhibited that one cannot derive special explanations of nature from the general forms of thought. Totality as a category leads to a conscious and unconscious work at forming observations into wholes. But where, as in the organic domain, phenomena already stand as wholes for observation itself, there the task must be to find by the road of analysis the laws that condition the subsistence of such wholes, not to appeal to the whole itself as a self-producing cause. On this point another philosophizing biologist already mentioned, J.

[17] *Mechanism, Life and Personality* (1913), p. 27.
[18] Driesch: The Problem of Individuality (1914), p. 41.

C. Haldane, stands more critically. He does not apply the concept of the whole causally, but sees the wholes that appear in nature as types of a higher form of reality (a higher or more concrete aspect of reality). On the other hand he thinks that where such a "higher" type appears, no causal problem is set at all: it is like a revelation of existence's essence, which does not interrupt the mechanical connection and does not deprive the physical-chemical concept of the world of its usefulness as a working hypothesis.[19] Like so many Hegelians, Haldane does not see that it is in reality a value-concept he is here operating with, in the same way as when one says, for example, that a work of art's aesthetic value is independent of the way in which it came to be. In the next chapter we shall come to discuss the relation between totality and value.

When it thus shows itself that the positive proof of vitalism is only a form of the negative, one asks with a certain curiosity how one is really to think of the new force that according to vitalism comes forward in the organic phenomena to fill the gaps that mechanical explanation of nature is supposed constantly to leave. Driesch informs us on this that it is not a spatial cause, although it works on (into) what is in space. If by cause one wants to understand spatial cause, then the ordering factor at work here, which Driesch would call entelechy, is not a cause. Where it displays itself as directing, not merely as forming, Driesch calls it a psychoid. The psychoid is not of a psychical kind, but is an analogue of the psychical, and Driesch finds vitalism's chief difficulty in this assumption of unconscious analogues of mental life. The psychoid is however to be not merely necessary for understanding certain biological phenomena, but is also to explain how it is possible that psychical being (consciousness) can arise.

For my part I must with respect to this last point remark that a distinction must indeed be made between the two things that in Driesch run together into one. I take it to be unavoidable to presuppose psychical analogues at lower stages of material existence, if one will not assume that conscious life arises with a leap, and if one will not undertake to prove its arising from purely material causes. But one has not therefore the right to let such an analogue appear as a *deus ex machina* in biology. —

A concept akin to vitalism is the concept of orthogenesis,[20] which has been set up to

[19] *Mechanism, Life and Personality*, p. 98, 137. — For Haldane the biological point of view is "higher" than the physical-chemical, the psychological again "higher" than the biological. Matter, organism and soul designate different stages of approximation to true reality (p. 117, 123).

[20] Bergson's "élan vital" is in Driesch's eyes too indeterminate an expression. He prefers the concept of

explain how development can advance from lower to higher forms. This explanation is then found in an original need or tendency to development in a determinate direction. There is in this connection no ground to go into this concept, since the criticism of it would be quite corresponding to the criticism of vitalism. The concept of orthogenesis is to the concept of development what the concept of vital force is to the concept of the organism as a whole. —

The result of our discussion is that the impossibility of a mechanical origin of life cannot be established — while yet the possibility that life has no absolute origin must not be lost from sight. If it should be possible to establish that life, the organism with its character of wholeness, can be explained by the mechanical road, the concept of the whole does not therefore lose its significance. On the contrary, the whole then shows itself precisely to have a deep ground in existence's elements, since these can in virtue of their own laws unite to form wholes. It is then the cooperating causal series that make such complex phenomena as organisms possible. The concept of the whole sets, like every category, a task, a question, but says in itself nothing about how this question is to be answered.

The special biological investigator need lay neither mechanicism nor vitalism at the base. It is his business to apply physical and chemical points of view within the organism as far as is possible at all. And any absolute certainty that he stands at the boundary of the possibilities he can never obtain. —

It has in this whole discussion been the chief task to bring out the general problem and to show its connection with the nature of human thought, as this displays itself in the different domains from which it fetches its subject-matters. It is easy enough to mock at vitalism in a more or less tasteful and witty way. But one serves thinking better by pointing clearly and definitely to the problem that here makes itself felt again and again, even if there are those who behave as though it did not exist.

21. The personality, the self, is another example of a whole that meets us immediately and is then made an object of analysis. The relation between the given whole and scientific investigation is here a counterpart to what made itself felt in the case of organic life. Here there appears an opposition between a conception that regards the personality

individuality, because here it is a question of a fundamental concept, a category. (*Science and Philosophy of the Organism*, II, p. 305, 313.) He finds moreover many points of contact between his conception and Bergson's (ib. p. 292 note). On Bergson's vitalism I refer to my book *Henri Bergson's Filosofi*, p. 35–42.

(and conscious life as a whole) as a mere result of an interplay of elements, and another conception that finds the explanation of the single phenomena of consciousness in the self, the personality as a whole. The first conception is represented most ably and most characteristically by the English "associationist psychology" (Hume, James Mill, Stuart Mill); the second is held not merely by the spiritualistic psychology, which lets the soul as a peculiar "substance" lie at the base of the phenomena of consciousness and intervene in their course, but also by more recent psychological conceptions that let the "I" or the "self" exercise a similar function. In this domain too it is the cognition of laws, so far as such is possible, that leads out beyond the contending conceptions. The only law that can be traced everywhere in the psychical domain is the law of synthesis. When the concept of synthesis is apprehended as a fundamental concept in psychology, what is meant is not the concept of a force, but precisely the concept of a law or a form for all conscious life. What psychological analysis can find will always show itself to be special forms of this fundamental form. It is not superfluous to remark that the question how far the fundamental law of synthesis can be exhibited everywhere in conscious life is independent of the answer to the question whether conscious life can be explained at all as a product of physical and chemical elements. Whether one answers this question with yes or no, psychology's own task remains the same; the relation between subject-matter and interpretation within psychology is not thereby altered. The very categories with which physical and chemical research works are special forms of synthesis, which like all other categories have developed under thought's striving to interpret the subject-matters presented. Thought itself is, from a psychological and from an epistemological point of view alike, its own last problem, however all other problems may have been answered.

The relation between the psychological and the epistemological way of looking at things has been discussed above (§§ 5–6). In the present connection we make psychology's position an object of epistemological investigation, just as psychology for its part investigates how it is possible that a scientific consciousness can develop. What epistemologically is the only road to truth is, psychologically regarded, only one of many possible directions conscious life can take, and it then becomes a matter of investigating how it stands to the other directions. When psychology is made an object for epistemology, it is in order to investigate the warrant of the road to truth that psychology as a science follows. And the most important question here is that of the character of wholeness of conscious life.

Both subjective and objective psychology — that is, both self-observation and observation of the outer phenomena to which conscious life is in one way or another assumed to be attached — lead to laying weight on conscious life as a little world whose elements stand in constant reciprocal action with one another according to determinate laws, and whose present is determined by its past. In the relation between past and present, that is, in the historical character of conscious life, the law of wholeness, the law of synthesis, displays itself especially plainly. This historical character now belongs, as we shall later see, to all material natural phenomena too. With them also the past must always be taken into account. It holds of organisms, planets and stellar systems, and it holds even of such lifeless things as manufactures. Two nails may look exactly alike; but if the one has been hammered several times, its molecular state will differ from that of the other, which has not undergone such treatment. Statics is in general, when it is isolated from dynamics, "an incomplete and unfruitful science, a branch severed from the trunk".[21] But in conscious life this historical character appears in the most significant way.

This historical character is given with the position of the concept of synthesis in psychology. The conscious and unconscious after-effects of the preceding elements and states are constantly connected with the following elements and states, and thereby the degree and constitution of the latter are determined. Fechner's law and the law of contrast point in this direction. This relation appears especially plainly in sensations of movement. If our own movement is to be sensed, the earlier impressions from the states and positions of muscle-fibres, joints and skin must constantly be combined with the following ones, and only by reason of this constant gathering-together do sensations of movement arise. What physiologically regarded is a successive manifold appears psychologically as a continuous whole (see my *Psykologi*[6] V A, 6, end). One has with right, in accordance with this, spoken of a "motor sensation of wholeness" as fundamental for the apprehension of ourselves as willing and acting persons.[22] As a counterpart to such elementary phenomena, in which nevertheless the fundamental law of conscious life displays itself, we may refer to the resolution — in the strictest sense of that word, as opposed to impulse and to purpose; — in it a decision is determined by all the elements

[21] Painlevé: *Mécanique* (De la Méthode dans les Sciences, 1909), p. 370, 400.
[22] Bror Gadelius, see above p. 25.

in the personality and by the personality's history in the past. In such a resolution both the real and the formal side of the personality, both content and form, both "soul" and "spirit", appear in a pronounced way.

When weight is often laid on self-consciousness, self-recognition, as a characteristic expression of the personality, it must be remarked that the self-identification these expressions designate precisely presupposes a comprehensive synthesis, within which an analysis, a special bringing-forward of the elements that condition the self-recognition, can take place. And the very elements that are identified in the self-recognition must be gathered together if this is to become possible. As KANT expresses it in his language, the analytic unity of apperception (i.e. the representation of consciousness's identity in a manifold of given representations) is possible only on the presupposition of its synthetic unity. One must remember, he says further, that consciousness is not a [single] representation, but a form of representing. We move here in a circle, since we must always make use of consciousness in order to be able to say anything about it.[23]

From another side too the concept of the whole appears in the psychological domain. The sharp differences between psychical states and between psychical elements — the differences that lie at the base of the distinctions and divisions that analytic psychology undertakes — appear clearly only during the more advanced development of conscious life. The more primitive states are more uniform and at the same time more concentrated. Development consists here, as in other domains, in advancing differentiation. To understand a decisive difference or opposition in the psychical domain one must therefore look back on the historical development that has led to it. The same holds naturally of the concentrations that can in turn (for example in resolution after deliberation) replace differentiations. But it is of the greatest importance to note this historical character, so that one shall come to lay neither too much nor too little weight on psychological divisions. Much criticism directed against the "threefold division" in psychology becomes objectless when the point of view here intimated (which is laid at the base of my presentation of psychology; see its Chap. IV) is held fast. I here leave aside the fact that one can use the threefold division schematically and yet undertake a reduction of the three kinds of element to one (for my part to the elements of will).

With the historical character of conscious life it hangs closely together that psycholog-

[23] *Kritik der reinen Vernunft*^{2}, p. 133; 404.

ical problems rest for a very great part on apparent interruptions of the direct connection in conscious life. In such cases it is a matter of finding a deeper-lying connection in virtue of the laws that have shown themselves to hold for conscious life. An important example of this is the doctrine of the displacement of motive, which is grounded in the influence that representations actually exercise on feelings; in this influence there lies a possibility that when a representation leads to another representation, the feelings attached to the first representation will also be able to become attached to the second and to remain so, even if the first representation falls away entirely. There is here a possibility of spiritual development, which Spinoza and the English psychologists have brought out.[24] In the domain of cognition there is a great example in the work of overcoming the hindrances to continuous thought-life that chaotic difference-series present.

There are however psychological problems of another kind as well, namely such as require us to understand the nature and coming to be of complex psychical phenomena. It is a matter of describing these special psychical wholes, analysing them and finding the conditions of their arising and subsisting. One can ask, for example, about the essence and conditions of religious faith, of conscience and of interest in art. In my *Religionsfilosofi* I have investigated religion from this point of view, and in "Den store Humor" I have given another example of such an investigation. —

I have hitherto kept essentially to subjective psychology, which rests on self-observation but can moreover very well make use of everything that can be elucidated by the experimental, physiological and historical road. A purely objective psychology would try to abstract entirely from self-observation and keep to the outer observations presented, especially of a physiological kind. It will of course rest on an illusion when one thinks by this road to arrive at a psychology, for without the acquaintance with conscious life that self-observation gives, one would not be able to know at all what the physiological phenomena one sees before one mean. Such an attempt has nevertheless been made by Bechterew in his "Objective Psychologie". I here leave entirely aside that a critical review of this work shows self-observation to be covertly present at many points. But what has interest in this connection is that Bechterew himself, from his purely physio-

[24] I take the opportunity to correct an error in "Den nyere Filosofis Historie", II, p. 372 f. The "Appendix B" there spoken of, in James Mill's "Fragment on Mackintosh", is a reprint of a chapter in Mackintosh. The credit for the clear presentation of the theory of the displacement of motive belongs accordingly not to James Mill but to Mackintosh, although James Mill in principle holds the same theory.

logical standpoint, thinks he can exhibit a “personal sphere” which comes forward in its independence especially when the organism’s state is determined not by the constitution of the actual impressions but by the “traces” of earlier impressions. The personal sphere is the sum-total of such traces, which constitute the organism’s “neuropsyche”, and whose reactions we call actions or deeds.[25] From a purely objective point of view too, then, the personality appears as a centralized, historically developed whole, which it is research’s task to analyse.

22. In the sociological domain, as in the biological and the psychological, a “mechanical” conception stands over against an “organic” one. The question is whether a whole — here a society or a state — can be explained merely by an outer connection of elements that in and for themselves could exist without such connection.

A purely external connection seems to be present when previously independent individuals enter into an agreement, a contract, in which they undertake to observe certain rules, so that a common life and a common activity become possible. It is accordingly as a kind of contract that mechanical sociology, as it appears in the old doctrine of natural law and in many forms of modern individualism, thinks of society’s arising. Opposition to an existing society crystallized in determinate forms often makes use of this way of looking at things. In sharp opposition to the contract theory the so-called historical school in the nineteenth century maintained an “organic” conception of society, according to which it appears as a whole — often a mystical whole — which, far from being a mere sum of independent individuals, precisely determines the individuals in their innermost nature and in their way of acting.

Hobbes is the most pronounced representative of the first conception, Comte and Hegel of the second.

When the first conception uses the point of view of a contract to explain society’s arising, it uses a point of view that has meaning only when a society and a legal order maintained by force already exist. What was to be established is accordingly presupposed. The old teachers of natural law already saw this, when they say that the society-founding contract is a tacit presupposition (a supposed covenant, as Hobbes expresses it), and so not entered into with clear consciousness but “lying in” the involuntary way in which the relation between men can arrange itself. Society and its ordering are accordingly a

[25] *Objective Psychologie.* Deutsche Übers. (1913), p. 432, 435.

fruit of the direction, or the interplay of directions, that life had already involuntarily taken among the individuals, before there could definitely be any question of social forms and institutions. The concept of direction presses forward here too. Just as it is an illusion when one speaks of atoms, forces or movements without taking account of the determinate directions in which atoms move and work, so it is also an illusion when one speaks of individuals without regard to the way in which life has led them to common life and common action with others, already before they can consciously arrange their relations. There has then never been a time when men lived in a pure "state of nature". This appears characteristically in HOBBES's proposition that in the state of nature there are no sons (filium in statu naturali intelligi non posse, De cive II, 10), since already the relation between mother and child, if it is to have legal or moral significance, must be seen under the point of view of a tacit contract that the child enters into by receiving the mother's help. It could not be better expressed that society is present in the first expressions of human life. HOBBES saw clearly too that in order to carry through his social mechanics he had to think of men as shooting up out of the earth like mushrooms, each by itself, and then afterwards entering into connections. The analogy HOBBES in reality rests on is that between individuals and states. Different states can indeed come to be quite independently of one another, and in the relation between them there can therefore be a question of a state of nature, which according to HOBBES is essentially a state of war.[26] He thinks of the relation between individuals on the model of the relation between sovereign states. But the question then returns when we turn our attention to the coming to be of each single one of these states.

Characteristic of the natural-law conception is that one wants to explain involuntary, partly unconscious social states by help of such social processes as go on with more or less clear consciousness in the single individuals taking part in them. It is a procedure analogous to the one that leads many psychologists to explain mental life's primitive and involuntary expressions by help of elements found by analysis of highly developed and differentiated states. —

The "organic" conception maintains that social life is always present in one form or another and to one extent or another, as soon as a human life exists at all. In the

[26]In all times, Kings, and Persons of Sovereign authority, because of their Independency, are in continual jealousies, and in the state and posture of gladiators, having their weapons pointing, and their eyes fixed on one another. *Leviathan*, I, Chap. 13.

primitive family, in any case in the relation between mother and child, and in life in the herd — phenomena occurring everywhere men live — a society is already given. On this conception society is the primary thing, and the individual, however independently it may develop, is socially conditioned and determined according to its whole nature and all its conditions of life. Every human expression of life and form of life finds its last explanation in social influence. On this conception the unconscious and the involuntary, everything that precedes clear consciousness, inference and calculation, comes into its own. It is seen that the single individuals do not arise or develop in isolation, but are what they are through the whole connection within which they have come to be. Here the whole is not a mechanical aggregate, but an original reality whose history is history proper.

The "organic" conception fits social life best in its more primitive forms, while the "mechanical" conception can appeal to experiences from later stages of social development. As Sir Henry Maine has shown in a series of significant presentations (first in "Ancient Law", 1861), the difference between more primitive and more developed forms of society can be designated as a difference between condition (*status*) and compact (*contractus*). Later this opposition has been more closely carried through by Spencer, Durkheim and Tönnies. Tönnies especially (Gemeinschaft und Gesellschaft, 1887) has shown that this opposition hangs together with oppositions between the involuntary character that the life of will has at its earlier stages, where it springs immediately out of nature (as "Wesenswille"), and the conscious willing conditioned by choice between determinate possibilities ("Willkür"). In opposition to instinct and involuntary devotion and adherence there now enter comparison and deliberation of ends and means, just as in society ties of family are replaced by agreement, custom and usage by legal rules and the conflict of interests, and just as faith is replaced by science. They are two social types of wholeness, which can recur at otherwise highly different stages of culture and replace one another in a rhythmical way.

But within both types, the contract-determined as well as the nature-determined, it holds that the individuals always belong to a whole, while on the other hand social life subsists in and through the single individuals, whose initiatives often become of decisive significance for social development.

Everywhere the concept of totality occurs, it holds that it cannot be used for the explanation of single determinate phenomena. Just as little as vital force explains any single organic phenomenon, and just as little as the concept of the soul explains any single psychical phenomenon, just as little does the concept of society explain any single social phenomenon. What matters, in the social as in the psychological and the biological domain, is to find the law, or the laws, through which a connection of the whole displays itself. There is a sociological mysticism, just as there is a psychological and a biological mysticism. I find such a mysticism, for example, in DURKHEIM's proposition that a fact is not social because it is found again in all society's individuals, but conversely: il se répète chez les individus parce qu'il s'impose à eux.[27] It is accordingly in the parts because it is in the whole, not conversely. But a social form must have developed before it can force itself on the individuals. And it develops only through the constant reciprocal action between individuals, a reciprocal action in which the single individuals are each of them both giving and receiving. At no point does individual independence disappear entirely. Every single individual is, after all, a little whole, as both biology and psychology must impress. The independence expresses itself both in the kind and degree of receptivity and through initiatives that can be formed through reciprocal action with other individuals. In the course of this reciprocal action the new contribution springing up in a single individual's interior can undergo alterations, partly through involuntary displacements, partly by conscious additions and combinations, partly by conflict with initiatives from other sides. GRIMM taught that folk-poetry had sprung from the soul of the people, but A. W. SCHLEGEL maintained against this mystical-poetic conception that it is due only to the circumstance that we do not know the authors of the single poems.[28] And BURCKHARDT has carried this further: "Folk-epic, folk-song and folk-melody seem not to be due to single great individuals; in place of these there enters a whole people, which we imagine *ad hoc* in a particularly happy-naive state of culture. But this substitution rests chiefly on the defectiveness of the tradition. The epic singer whose name we no longer know (or know only as a collective name) was great in the moment when he for the first time gave a branch of his people's cycle of legend a lasting form; in that moment the folk-spirit concentrated itself in his in a magical way… The concentrated folk-spirit speaks out of him; otherwise the song would not live through the ages".[29] — Everything new must stem

[27] *Les règles de la méthode sociologique*, p. 14.
[28] C. T. GOOCH: *History and Historians of the nineteenth century* (1913), p. 57.
[29] JACOB BURCKHARDT: *Weltgeschichtliche Betrachtungen* (1905), p. 224 f.

from single persons, in the last resort from a single individual in whose consciousness a peculiar crystallization has taken place. It then becomes a new task to investigate how this crystallization has been able to take place, and here one will again stand before the reciprocal action between individuals and between them and the existing traditions. It is mythology when the historical school (especially the Prussian) regards the single individuals as merely secondary. The new beginnings can, before they lead to social organization, be studied only in the consciousness of single individuals, so far as it is accessible. Here mutations of highly different lifespan can arise. Here a constant conflict goes on between the new and what is handed down, already in the single mind. Such processes cannot be understood when one makes society into a mystically working force and holds a social vitalism.

An interesting example of a conflict of this kind is Erik Gustaf Geijer's relation to the historical-romantic school to which he had enthusiastically belonged. By his own account it was history itself that led him away from the historical school. History showed him the significance of individual initiatives. He came to see the possibility that a generation or a group of men, in confidence in the future and its possibilities, could so to speak renounce inheritance and debt (*gøre sig urarfva*).[30] Tradition is pushed aside, new paths are struck out, and there comes an apparent break in the continuity. But all such apparent breaks only invite the investigator to find a deeper-lying connection than the one at which one had hitherto stopped. A leap or a break means for science only a task, whether this can be solved or not. It has been said with right of Alexis De Tocqueville that in his book "L'ancien Régime" he did for history what Lyell did for geology, in refuting the catastrophe theory. The Revolution now came not as a miracle, but stands as prepared through the absolute monarchy's centuries-long tendency to concentrate all power at a single point; the revolutionary government was only a continuation of this concentration, and the Revolution became possible because it was only a matter of pressing in upon that single point.

The more a deeper-lying connection is found, the more difficult it becomes to divide history into periods. The customary division of Europe's history into antiquity, the Middle Ages and modern times is said to stem from the philologists of the seventeenth

[30] *Samlade Skrifter*, I, 7, p. 137. — I was made aware of this passage through Dr. Hilma Borelius' treatise on Geijer before "Affaldet". — Cf. moreover *Personlighetsprincipen i Filosofin*, p. 39–49, where I discuss the problem in general.

century, for whom the centuries preceding humanism came to stand as a completed period.[31] For the research of a later time the Middle Ages may seem to dissolve into a whole series of renaissances from the ninth to the fifteenth century, and it may become a question whether we are not living in the Middle Ages still.[32] And an eminent historian has even declared that he does not acknowledge the opposition between the history of antiquity and that of modern times.[33]

23. In the sociological domain too, then, the concept of totality contains in itself no explanation. It is a matter of finding the law that holds the elements together and in virtue of which the whole subsists.

Sociology is a historical science. Social life's future we do not know; its present we cannot survey; it is then social life's development in the past that is the given, the subject-matter. Historical research aims at the description and understanding of events, institutions and personalities. It applies here the same method as natural science. Its criterion of reality is the same. The past is not immediately given, only the report of it or monuments from it; but neither does the natural scientist have nature immediately given, only observations of natural phenomena. Just as the natural scientist has, by comparative observations of natural phenomena, to form for himself a whole-image of a natural connection, so the historian has, through comparison of reports and monuments, to form for himself a coherent conception of the past. Everywhere it is the lawlike connection that is the mark of reality. Just as contradiction between observations is got out of the way by exhibiting that the different observations are due to different points of view, sense-organs or means of observation, so that what from one point of view appears in a certain way must from another point of view consistently appear in another way, — so contradiction between different reports can be explained by the same event having had to be apprehended in different ways by different reporters according to their presuppositions. The English historian Gardiner has had a special capacity for seeing the significance in which historical personalities had to come to stand for their opponents.[34] It is not enough to correct observations or reports; it must also be shown how the observations and reports that need correction have been able to arise. Perfect truth must be able to

[31] Joh. Steenstrup: *Historieskrivningen* (1915), p. 150.
[32] François Picavet: *Histoire comparée des philosophies médiévales*, Chap. 4.
[33] Freeman: *Methods of Historical Study* (1886), p. 20.
[34] Gooch: *History and Historians*, p. 362 f.

show how what appears in a certain way for one subject must appear in another way for another subject — whereby the investigator must naturally not forget that he is himself a subject too. Every definition of what truth is has significance only by setting the task of finding new subject-matters and new points of view and an ever closer connection both between the points of view and between the subject-matters.[35]

There is in this respect only a difference of degree between natural science and history — a difference of degree resting on the exactness and completeness with which the criterion of reality can be applied in detail. For the historical investigator the exact method the natural scientist can apply does not stand at his disposal. But that conditions no difference of principle.

On the other hand a difference of kind between natural research and historical research has been thought to lie in this, that natural science aims at general laws, whereas the goal of historical science is the apprehension and understanding of individual phenomena (events, institutions, personalities). HEINRICH RICKERT has in a very interesting way made this difference between history (cultural science) and natural science prevail. He declares expressly that he hereby lays the weight on the ends the different sciences set themselves, not on the means they work with.[36]

Is it then now natural science's last goal to exhibit general laws for what happens in the world? The interest in finding general laws has naturally developed out of the need to distinguish as exactly as possible between reality and imagination, since the criterion of reality is precisely the fixed, unbreakable connection between different subject-matters. But the finding-out of general laws has then become an object of independent interest; there can be an interest in finding general laws even if the insight into them acquires no practical significance. The single investigator can stop at a law and see his life's task in exhibiting and confirming it. But the more laws are found, the more there is revealed to us a great whole that we call nature, and in which the interplay of the single laws (or as H. C. ØRSTED said: *Samfoldighed*) is what makes the whole a whole. Nature in this sense is a *unicum*. Existence is only one, and it constitutes an individual whole, since individuality according to LEIBNIZ's definition consists in a law of continuity through the variations. When RICKERT says that the single and individual interests the natural

[35]In *Den menneskelige Tanke*, p. 303–305, I have sought to show how this conception is reconcilable with the proposition that truth can only be one.

[36]*Die Grenzen der naturwissenschaftlichen Begriffsbildung*^{2}, p. 302.

scientist only so far as it can contribute to understanding the material world as a whole (p. 60), he himself concedes that laws are in the last resort to serve for understanding the great individuality we call nature: Totam Naturam unum esse individuum (SPINOZA: Ethica II, Lemma 7). Natural research accordingly studies existence's character, or its biography. Historical research studies the biography of mankind, especially of European mankind. A higher task, which perhaps lies beyond the domain of cognition, would be to gather these two biographies into one. But however that may be — if there is to be any talk at all of a last goal for science, then this goal falls under the concept of totality, not under the abstract concept of the general.

RICKERT will answer to this that it is the single, limited individualities he has in mind. History seeks to characterize single events or personalities that are unique and do not recur; it has to do with "das Einmalige". Natural science on the other hand seeks in its numbers and formulae to determine the relations that constantly repeat themselves. The single, individual phenomena are for the natural scientist only examples (p. 354 f.). Against this I have urged, in a criticism of RICKERT,[37] that absolute repetition does not prevail in material nature either, if only one analyses the single cases; "das Einmalige" rules both in nature and in history. In the new edition of his work (1913) (p. 468) RICKERT concedes in his reply to my objection: "Es findet in der Natur wie in der Geschichte jede Begebenheit nur einmal statt, und es gibt in Wirklichkeit keine Wiederholungen". I cannot see otherwise than that the difference of principle between natural science and history thereby falls away. The goal must after all be to understand the subject-matters, and when the subject-matters are always individual, cognition of the individual must be the goal for every science. After the general laws have been found there will then always remain the task of explaining whence comes what is not repeated. "Das Einmalige" always sets a problem, whether this can be solved or not. It shows itself too that natural science sets itself the task of exhibiting and explaining special problems of development (of stellar systems, planets, mountains, rivers, organisms, species), and it is a desperate expedient that RICKERT seizes (p. 449, 454) when he declares that the doctrine of development belongs to history's domain and not to natural science's.[38] The

[37] *Göttinger gelehrte Anzeigen*, 1906. Cf. *Den menneskelige Tanke*, p. 214 f.; 224 f. — My criticism concerned the first edition of RICKERT's book.

[38] Conversely he reckons sociology to natural science (p. 611). But sociology is, as TÖNNIES (*Soziologie und Geschichte*, Die Geisteswissenschaften, 1. Jahrg.) shows, a comparative historical research. It is as

different hypotheses of development are precisely attempts to understand the coming to be and subsisting of individual wholes by help of the laws of natural science. It is a similar task the historian sets himself when he investigates the conditions of a historical phenomenon's occurrence.

Georg Simmel, whom Rickert (p. 269) regards as his predecessor, teaches precisely that the total connection which conditions historical understanding stands above the opposition between the general and the individual, which one has hitherto regarded as an insuperable either-or.[39] Simmel's standpoint would have come more fully into its own if he had within the individual again distinguished between the typical (the characterization) and the concrete (the single expressions and situations).

Science's goal is neither the general nor the individual, but it is to understand reality, which is always individual, by help of the general laws — those laws which already decide what is reality and what is dream, and which at the same time yield the only possibility of understanding our dreams themselves. Our most perfect thoughts are neither general concepts nor concrete individual concepts, but typical individual concepts.

the result of such a comparative science that Tönnies's work *Gemeinschaft und Gesellschaft* is to be apprehended. It establishes a typical opposition, not a general law, as Rickert thinks (p. 469).

[39] Georg Simmel: *Die Probleme der Geschichtsphilosophie.* Eine erkenntnisstheoretische Studie, 2. Aufl. (1905), p. 34.

Chapter V

Totality and Value

24. There is in the present day a philosophical tendency that would make the whole of philosophy into a doctrine of value. Philosophy's field of work then consists of the problems of value, the questions of truth, of beauty, of goodness, of holiness. It starts out from the fact that these ideas make a claim to validity, and its task is then to investigate wherein this validity consists and how far it can be justified. The mere establishing of a fact already rests on a valuation, namely on the presupposition of truth's value. Truth is itself a value that demands acknowledgement. Theoretical thinking, which wills cognition, is to be regarded as a special case of a striving to realize values. If the validity of the value of truth is rejected, no judgment becomes possible. In science too, accordingly, it is a question not merely of thoughts and representations but also, and in the last resort, of values and valuations.

This way of looking at things has been made to prevail most fully and energetically by Heinrich Rickert in the work already mentioned on the limits of concept- formation in natural science. (See especially p. 587–613.) I agree that the point of view of value is not a wholly new point of view that first comes forward in economics and aesthetics, in ethics and the philosophy of religion; it makes itself felt at bottom, from the first and uninterruptedly, in all thoughts as well as in all wishes. In my presentation of human thought-life I have taken the road of presupposing, after establishing that the three elements subject-matter, form and value make themselves felt in every act of thought, an intellectual interest, an acknowledgement of truth's value, as the constant presupposition, and then investigating the forms of thought that this presupposition leads one to apply.

But this investigation then runs on its way into the concept of value itself — not merely because it forms the constant presupposition of the search for truth, but also because there are other values than the intellectual. Therefore the concept of value must be taken up in its generality as a fundamental concept, a category. The question is whether the quality of value can become an object of rational treatment, as the quality of sense and the quality of existence have shown themselves able to be.[1]

As with all qualities, we begin apparently in a purely subjective way. Value-judgments spring from pleasure and displeasure, from satisfaction and want. But pleasure and displeasure are symptoms of a need, a movement or a tendency in a certain direction, and they arise when the tendency is either favoured, so that it can be continued with full force, or hindered and halted. We run here again into the concept of direction. Whatever it is that we observe, it is always in movement in one direction or another. But when there is a question of values, it is presupposed that there is a whole with a tendency to assert itself over against the surrounding world, and it then depends on whether this tendency is furthered or hindered. And in the strictest sense there can be a question of values only from the standpoint of personal beings, because only there are the tendency and its conflict with circumstances noticed and felt. Only by the road of analogy is the concept of value carried over to other wholes (societies, organisms, inorganic wholes); it is indeed already analogy when a personal being seeks to enter into what can have value for other personal beings. A tendency that is noticed we call a need. There will accordingly be an inner relation between the concepts need and value. When what the need aims at becomes an object of consciousness in advance of the satisfaction, the need becomes an impulse.[2]

[1] *Den menneskelige Tanke*, p. 238 f.

[2] On the relation between need and impulse see my *Psykologi*[6], VI B, 2 c; C, 1. — VII B, 1 a. — It is Hobbes and Spinoza to whom we owe the plain distinction between need and impulse. Hobbes especially has put need in connection with a beginning movement in a certain direction. He is guided here by an analogy with Galilei's concept of force (momento, impeto), which finds application whether a body is at rest or in movement. Cf. *Elements of Law*, I, 7, 2 (endeavour, i.e. internal beginning of animal motion). *De Corpore*, XXV, 12 (the tendency to unhindered motus vitalis becomes, when experience and memory are possible, need, appetitus). A purely physical definition of conatus (as motus per spatium et tempus minus qvam datur) is given in De Corpore, XV, 2. (On this last point see *Den nyere Filosofis Historie*[2], I, p. 174. Tönnies: *Thomas Hobbes*[2], p. 116 f.) — For Spinoza reference is made to *Ethica*, III, 6–9, where a distinction is drawn between the tendency (conatus) to self-preservation, the need (appetitus) and the impulse (cupiditas), the last of which presupposes consciousness of the goal. In more recent literature the same difference appears in Stout (impulse — desire), Fouillée (appétit — désir) and others. In German literature *Trieb* often means both, not to the advantage of clarity.

An objective doctrine of value becomes possible through need and value showing themselves to hang together with a personal being's subsistence as a whole. They show themselves to vary with the conditions of this subsistence. An investigation of these conditions elucidates the arising and subsisting of values, and it makes it possible to distinguish between genuine and false values, grounded or ungrounded need, since it will be able to show whether the need that is felt, and the valuation it leads to, correspond to the actually given or possible conditions of life.

The concepts of value can be divided according to different points of view. In the formal respect a distinction can be drawn between elementary (sense-determined) and ideal (representation-determined) values, between immediate and mediate values (ends and means), between potential and actual values. In the real respect a distinction can be drawn, when the content is laid at the base, between biological, intellectual, aesthetic, ethical and religious values, and when the extent is laid at the base, between individual, social and cosmic values. How far a rational doctrine of value can be carried through for all these values is a matter by itself, and the question belongs under the treatment of the problem of valuation. In this connection we can keep to the real values, especially from the point of view of content, in order to elucidate the connection between the concept of value and the concept of totality.

25. That the different real values, as regards extent, presuppose certain wholes with a tendency to assert themselves, is evident. Every single individual, every social group and every people values the circumstances outside itself and within itself according to whether the conditions — the actual or the supposed conditions — for subsistence are present or not. But from the point of view of content the same can be shown. Economic life, intellectual life, aesthetic life, ethical life and religious life — these are so many different wholes of thoughts and endeavours, each with a tendency to close itself off in itself, and each determined by a ruling end with which the domain of life in question stands or falls.

a. The struggle for the maintenance of physical life can make the economic (biological) interests everything. There is here a side of life that has its own laws and in virtue of them marks itself off as a whole, even if this whole can in turn enter as a part into a more comprehensive whole of life. There is here also in fact a science of its own with its own points of view and methods. (See further Etik III, 14.)

b. That science is an example of whole-formation has been attempted to be established in preceding sections of this treatise. Scientific work aims at building up a world of thought within which every subject-matter can find its place. Intellectual values depend on what furthers this work and carries it in the direction of its goal.

c. Aesthetic value a subject-matter acquires by appearing as a characteristic whole and being able to make itself felt as such for immediate intuiting, apart from intellectual and practical interest. Since reality does not always present characteristic, fully individualized subject-matters, imagination can step in and give it the rounding-off without which a whole does not come to be. It is a relation that can be studied especially where we know the subject-matters (the motifs) beforehand and can compare them with what the poet or the artist adds through his work. Examples of this we have in the Greek tragedians' relation to the history of legend and in SHAKESPEARE's relation to his predecessors (as this is elucidated in SCHÜCK's book on Shakespeare). It is the image of a peculiar, individual whole that is aimed at. In applied art the impression of beauty is conditioned by the relation between the object's contour and the ornament's contour (between "contour-line" and "ornament-line"). For beauty to be attained they must either coincide, or else their points of intersection must lie in a simple rhythm — but this is also the condition for a character of wholeness to arise. So too it is the simply rhythmical in verse, as opposed to prose, that makes it appear with a greater effect of wholeness, and in this must be sought the ground of the greater aesthetic value that verse in its limitation can have.[3]

In philosophical systems there often prevails an aesthetic need for harmonious or rhythmical connection, a need that springs naturally from the theoretical need to exhibit connection in everything, even in what is most contradictory. Recently MARK BALDWIN (Genetic Theory of Reality, 1915) has maintained that a world-view that would avoid, or lead out beyond, all relativity must acquire an aesthetic or an artistic character; it must become "Pancalism". Otherwise it acquires a purely formal character, as in those systems where the relation between unity and manifold, identity and difference, plays the chief part. BALDWIN's train of thought recalls SCHELLING's "System des transcendentalen Idealismus" (1800), where artistic intuiting is proclaimed as the only possibility of getting beyond life's oppositions.[4]

[3]This observation I owe to a young friend who has occupied himself especially with ornament.
[4]Den nyere Filosofis Historie[2], II, p. 156.

d. All ethical valuation starts out from an interest in a whole whose subsistence and unfolding are to be asserted and furthered. Here the point of view of extent clearly comes forward again, since the conflict between ethical systems is conditioned by its being different, more or less comprehensive wholes that lie at the base. It may be the single individual's, or the family's, or the estate's, or the nation's, or humanity's subsistence that conditions the valuation in the particular case. In the third chapter of my Etik I have investigated these oppositions and their significance for the scientific character of ethics.

Sometimes one thinks that a valuation becomes objective merely by not being individual but social. Hence the frequent assertion that a scientific ethics must of course be a social ethics. Heinrich Maier[5] would limit the possibility of objective valuation to such cases as have a "relation to man as *genus*" at their base. But in itself a valuation from a more limited whole can be just as consistent and just as objectively grounded as a valuation from the most comprehensive whole we can imagine. The egoist can, for example, be fully objectively right that this or that conduct on his part is a necessary condition for the attainment of his end. There is always in an ethical judgment one of the premises that is determined by the relation to the determinate whole whose preservation and development are at stake. And it is this premise that causes the problem; for there is no logical necessity that a certain determinate whole should determine the valuation.

A comparative ethics will have to investigate how far the valuations made from the relation to different wholes as foundation can lead to results that agree (apart from the motivation). By such an investigation it will show itself that there are certain ethical categories that are applied at all ethical standpoints, even if they receive different places. Such concepts are duty, virtue, good, right. They all express one relation between a single action and the whole from which it is valued. In duty it is expressed that the whole's demand determines the action; in virtue, that the action springs from an immediate feeling of unity with the whole; a good is everything that serves the whole's subsistence; a right expresses that the action is not conditioned by the whole's demand, but does not conflict with it either.[6]

e. The philosophy of religion leads to a corresponding conception. All religion is conditioned and borne by a need to hold fast the constant possibility that the whole to

[5] *Psychologie des emotionalen Denkens*, p. 652, cf. 654–673.
[6] Cf. *Den menneskelige Tanke*, p. 350–361.

which man feels himself attached, or which he believes himself to be, can subsist and develop. That is what is meant when religion is said to be faith in the subsistence of value. It is always a kingdom, a greater or a smaller one, that man asserts in his religion.[7]

26. Everywhere subject-matters with a character of wholeness appear in our experience, we can have an interest — in any case an intellectual interest — in investigating their origin and the conditions of their subsistence and development. It can often be difficult here to mark off the purely intellectual interests from other interests. Aesthetically we can thus "feel ourselves into" or "live ourselves into" natural objects or historical personalities and events that have no practical interest for us. Since all value is attached to a whole as its presupposition, it can also naturally happen that any whole whatever appears to us as a centre of value, since by the road of analogy it acquires a gleam of the interest belonging to the whole we ourselves are or belong to.

In the modern theories of development, as they have appeared in the last two centuries, this point of view makes itself plainly felt. So far as man's position in the universe is determined by what fate organic life, the earth and the solar system have had and have, it is not merely analogy that conditions the interest in biological, geological and astronomical development. But no such interest attaches to the history of a single star or a single mountain, or to the geology of a district, when we do not ourselves live within their bounds. And it stands in a similar way with the interest in other times and peoples, and in personalities who have lived under quite other conditions than ours. RICKERT has here set up the interesting concept "historical centre", a concept that stands so to speak on the transition between the purely theoretical concept of totality and the concept of value. Historical research chooses such subject-matters as present the character of a whole, in that they stand both as a gathering-place and as a starting-point for whole series of events. Such a whole — be it a single personality, a people, an institution, a culture — stands as valuable in the light of what is gathered and concluded in it, and what proceeds from it, even if it has no practical influence, either in its causes or in its effects, on the whole the historian himself is or lives in. The historian must be able to enter into what, given a certain whole — a certain time, a certain people, certain tasks — acquires value, and it is this that determines for him the relation between the single parts of the presentation he gives. It is no valuation proper that the historian gives (as

[7] Cf., besides my *Religionsfilosofi*, *Den menneskelige Tanke*, p. 375 f.

historian), but at the base of his consideration there lies a relation to values (Beziehung auf Werte) which need not be his own, but into which he can enter.[8]

If this relation to values is entirely lacking, historical interest ceases, and we come then to what another acute philosopher of history, Georg Simmel, has called "the historical threshold".[9] This threshold can naturally shift, be lowered or raised, like Herbart's and Fechner's psychological threshold, according as the historical material makes it possible to adopt points of view of wholeness or not.

A third historical concept hangs closely together with the two named, namely the concept of historical greatness. This does not coincide with moral greatness, but is determined by the extent and reach of the forces working to produce a new whole in the national or cultural respect. However violent and selfish a historical personality may show itself to be, there will nevertheless — in real greatness — be what Burckhardt calls "a mysterious coincidence between the individual's egoism and the whole's use or greatness". Alexander did not make his conquests and found his world-empire for the sake of humanity and culture, but his deed nevertheless led out beyond the boundaries that differences of nationality had hitherto marked out in these respects. Great men are according to Burckhardt necessary, so that "the world-historical movement can periodically free itself with a jerk from extinct forms of life and reflecting chatter".[10]

The different historical schools would often disagree about where the historical centres lie, where the historical threshold is to be set, and how far the concept of historical greatness is to find application in the particular case. There may be a tendency to side with the vanquished, or to side with the victorious; weight may be laid on culture or on the state, on the spontaneously working forces or on organization through the application of power. Under all these oppositions it will always show itself to be the relation between wholes and the conditions of their subsistence that lies at the base of the different ways of looking at things. And by the road of analogy these oppositions will in the last resort — but also only in the last resort — be able to be led back to oppositions between ethical standpoints.

27. Through the concept of totality the concept of value points back to natural

[8] *Die Grenzen der naturwiss. Begriffsbildung*[2], p. 325–329.

[9] *Probleme der Geschichtsphilosophie*[2], p. 131.

[10] *Weltgeschichtliche Betrachtungen*, p. 245, 252. — Cf. Hegel's brilliant development of "the cunning of world-reason" in using human passions, in the introduction to his Philosophy of History.

science and historical research. For if there had not been in existence, as it once for all is constituted, a possibility of wholes arising — if accordingly all causal series had been divergent or parallel — then the concept of value could not have arisen. It is a contribution to existence's characterization that totalities can arise and subsist within it.

How the totalities whose existence value presupposes have arisen need not be decisive for the value itself. Value is a fact that is not overturned by the way in which it has come to be, just as little as a man's value is diminished by his forefathers' bad conduct. Even so unprejudiced a man as Guyau thought nevertheless that something valuable cannot arise from something that has no value or even has the opposite of value. It is a theological train of thought that makes itself felt here, and that would consistently have to lead to all research into the coming to be of values being forbidden in value's own name, just as Lohengrin forbids Elsa to ask about his name and descent. But it is not easy to check the need to ask and to inquire, the need that Wagner calls "Wissens Sorge tragen".[11] Nor is there any ground for misgiving. Values arise according to the great laws that make whole-formations possible under certain conditions, and precisely thereby, as already remarked, the concept of value hangs together with scientific cognition and can itself become the content of science.

But the world of reality is more comprehensive than the world of value. Here there are experiences to be made that can favour a feeling of humour in face of the world as it is — partly because values are often realized by roads to which we cannot ascribe value in and for themselves, partly because the world of value appears as an oasis within a desert.[12] Even if one could think of a valuable goal for reality's unsurveyable causal series, we should nevertheless not be able to know this goal, and we should in our valuation of existence have to keep to the parts and the stretches we can survey. The conception that everything in nature and history leads in the direction of a completion, a harmonization, a realization of all ideals, is due to chiliastic representations and optimistic postulates. Such a naive teleology has no firm foothold. If one rests on experience, one must rather grant Burckhardt that in history as in nature enormous preparations and a

[11] Nie sollst du mich befragen,
noch Wissens Sorge tragen,
woher ich kam der Fahrt,
noch wie mein Nam' und Art.

[12] *Den store Humor*, p. 70–73.

disproportionate din are employed with small results at the end.[13] There is prodigality in history as in nature. But neither does the concept of value as a category have teleological considerations as its consequence. Just as little as any other category can the concept of value be used as the sole fundamental concept from which all other forms of thought could be derived in their relative warrant. It is even the most concrete and complex of our categories, presupposing the others without being presupposed by them.

The opposition between human values and the world of reality came to clear consciousness with the new picture of the world that Copernicus gave the impulse to. The problem was set as sharply as never before: life is vanishing in relation to the mass of the earth, and the earth in relation to the world of the stars. Even if life conquered matter, so that the "élan de vie" of which a brilliant philosopher speaks could always hold its own against the tendency to distribution and equilibrium, the disharmonies begin again within the world of life, since the one whole of life subsists only at the other's expense, as the wolf over against the lamb. If every whole of life is to designate a natural end, there is strife within the world of values itself. And if instead of teleology one speaks merely of orthogenesis (§ 20), then experience shows that development can run itself fast in helpless, unviable forms.

If it could even be shown that all collisions between wholes, or between wholes and elements, were fruitful, served to carry development further, the concept of value would in spite of everything be able to assert a concluding position in our world of thought. But is not Goethe right against Hegel when he maintains that there is an absolute tragedy, and not merely conflicts through which truth and right come forward with new clarity?

Our view of the world and of life must, if it will preserve its connection with rational thinking, go other roads than teleology. Whether such roads exist the doctrine of categories cannot decide by itself alone.

[13] *Weltgeschichtliche Betrachtungen*, p. 183.

Chapter VI

Totality as a Limiting Concept

28. The expression *limiting concept* was introduced by KANT to designate a concept we necessarily arrive at but cannot carry out, cannot give positive significance. Such a concept KANT found in the concept "the thing in itself". Since among the presuppositions of our cognition there is necessarily a subjective factor which displays itself in the fundamental forms of thinking, we cannot cognize existence as it is in itself, and yet we must, KANT thinks, necessarily assume that existence has a constitution quite apart from the way in which we, according to the nature of our thinking, apprehend it. This assertion was, as is well known, a great stumbling-block for KANT's nearest successors. It would indeed be remarkable if cognition should, in its own logical advance, lead to a point where it should itself no longer hold. Such a point KANT found in the question of the ground of the subject-matters (the "material") in working up which human thought has use for its forms, its presuppositions. But — precisely according to KANT's own philosophy such a question cannot be raised! For according to "Kritik der reinen Vernunft" thought has no other task than that of working up given subject-matters and explaining the one subject-matter by the other. Where no subject-matter is given there can only be formal science (logic and mathematics), which precisely abstracts from all special subject-matters and investigates the pure forms of cognition. In real science (natural science, psychology and history) subject-matters are worked up so that the rational connection KANT calls "experience" becomes possible. Neither formal nor real science runs into the thing in itself — the former because it presupposes no "thing" at all, the latter precisely because it always presupposes "things" as *given*.

But KANT has another limiting concept besides "the thing in itself", and that is the concept of absolute totality. This concept designates something that cannot be a subject-matter for us, since every subject-matter is limited, every given whole stands over against a surrounding world with which it stands in reciprocal action. The concept of absolute totality has according to KANT the significance of pointing out beyond every boundary one might be inclined to stop at — of referring to new tasks every time the work with the subject-matters hitherto presented seems to be completed. This concept too we arrive at with necessity, according to KANT, since the fundamental form of our thought is the synthesis, which always demands that what has hitherto been gathered be gathered together with new elements.

On closer consideration it now shows itself that the boundary lying in the concept of absolute totality (in what KANT calls "Ideas") coincides with the boundary the concept "thing in itself" was to indicate. Where KANT criticizes the dogmatic concept of "the world" as a given absolute whole, he declares: "In the empirical regress no experience of an absolute boundary can be met with… The ground of this is that such an experience would have to contain a limitation of an appearance by nothing,… which is impossible".[1] This declaration strikes "the thing in itself" (which can never become an appearance) as well as absolute totality (which can never be presented as given). And it is clear that we could come to "the thing in itself" only if our cognition were completed, formed an absolute totality; only then could we come to ask about the origin of the subject-matters we had worked with; until then we had enough to do in working together the new subject-matters that presented themselves. But then we stand precisely at the other boundary, at the ideal that the concept of absolute totality sets, and sets at every stage of our cognition's development. The task is always the same: beyond every given whole to seek a greater connection, a greater whole, and within every given whole to seek back to simpler elements, which will then in turn show themselves to be wholes in miniature.

In KANT's "antitheses", on which he vainly tries to foist a dogmatic train of thought, his own philosophy is in reality contained. In the "antitheses" there is operation with KANT's own central concept of "possible experience", and he says expressly that they express the purely scientific standpoint.[2] They make no absolutely completed system

[1] *Kritik der reinen Vernunft*[1], p. 517.
[2] *Kritik der reinen Vernunft*[1], p. 468 f.

possible; but that is indeed also the real consequence of KANT's philosophy. So long as thought is awake, a constant conflict takes place between a given and the tasks it sets. Cognition works always at coming to constitute a perfect whole, but so long as the relation between subject-matters and form repeats itself, the task presents itself anew. —

As has been exhibited in the preceding sections of this treatise, the concept of totality is not merely a special category having its determinate place in the series of categories: it expresses at the same time a peculiarity of every one of our forms of thought. Therefore KANT's sharp distinction between category and Idea cannot be maintained either.[3]

29. If we now look more closely at the concept of totality as a limiting concept, it presents — as we must also expect in the case of a limiting concept — partly a negative, partly a positive side.

a. The negative side, which arises from every whole — whole of thought or whole of subject-matter — being determined by its relation to something other than itself, gives occasion in turn both to a formal (logical) and to a real (empirical) consideration.

α. No concept can be determined except by its relation to other concepts, no judgment grounded except by its relation to other judgments. Every concept and every judgment is a point where the one leg of thought's compasses is set, so that a point may then be sought where the other can be set. This is what leads to setting up relation (or synthesis) as the first member in the series of categories. Therefore an absolute conclusion would conflict with thought's own laws. It ends in any case with a problem, when the other leg of the compasses can find no point.

When speculative systems taught a conclusion in the thought of something that "subsists in itself and is understood through itself", that was a transition from thinking to mysticism. Nothing subsists through itself, and nothing is understood through itself. Speculative philosophers, from PLATO and ARISTOTLE onwards, give us the choice between acknowledging something that is understood through itself and subsists through itself, and ending in scepticism. Thus SPINOZA says (Den korte Traktat I, 7, 9) that when one cannot cognize what lies at the base of everything, one cannot cognize the single things either, which have their ground in it. Even in the most recent time such an ontological train of thought comes forward. BERTRAND RUSSELL lays it at the base of his criticism of the conception according to which the criterion of truth is the mutual connection

[3] *Kontinuiteten i* KANT's *filosofiske Udviklingsgang*, p. 29–35.

between our thoughts and observations. The result of this conception is scepticism, RUSSELL asserts: for a perfect and all-embracing connection cannot be exhibited, and we could therefore, according to that criterion of truth, not be sure of the validity of a single judgment.[4] But the conception RUSSELL combats says at bottom only that truth must be won by constant work, and that the truths we already believe ourselves to have won must always be tested anew through their relation to new thoughts and observations. Here is a work that cannot stop. And the continued work rests on the constant presupposition that truth must constitute a whole. Even the surest results that have been obtained show themselves to be limited and to need supplementation. An ever more intimate connection can be obtained between an ever greater sum-total of observations. But grounded assurance can never be obtained that one is now at "something that subsists in itself and is understood through itself". And — even if it were possible to get so far, truth would then also consist in a connection, namely connection with what subsists in itself, and there could then be a possibility of considerable doubt when this connection had to stand its test in detail.

β. So far the formal consideration. But what holds for all thoughts and observations holds also for all subject-matters. Every subject-matter that can be an object of observation and of thinking elaboration shows itself to be subject to determinate conditions and bound to determinate boundaries, so that in order to understand it one must take account of the whole connection in which it occurs. Every subject-matter stands in relation to a surrounding world that presents either support or hindrance for its subsistence. The more science passes over to applying the insight into the general laws to understanding individual wholes and finding the conditions of their subsistence, the more will this way of looking at things show itself in its warrant and in its significance. Right up into the highest forms of human spiritual life this investigation can find tasks. —

The root of the problem of cognition lies in the circumstance that truth, both as real and as formal truth, must constitute a whole. Instead of the old opposition between "the absolute" and "the relative" one must now, as has rightly been said, put the opposition between the total and the partial.[5]

b. The positive side of the investigation of a whole concerns in the first place the

[4] *On the nature of truth* (Proceedings of the Aristotelian Society, 1906–07). Cf. above § 18 a.
[5] LÉON BRUNSCHVICG: *Les Étapes de la Philosophie mathématique* (1912), p. 414.

relation between unity and manifold within it. The unity displays itself in the thoroughgoing lawfulness according to which the parts belong together — the manifold in the elements that are connected in virtue of this law, but each of which can possess properties not determined by it alone. As has shown itself in the foregoing, the different wholes present great inequalities as regards the relation between unity and manifold. And the philosophical systems exhibit here one of their most characteristic differences, according as the weight is laid on the binding tie or on the manifold elements it connects. In the history of philosophy this difference comes plainly forward when one compares Parmenides with Democritus, Plato with Aristotle, Spinoza with Leibniz, Hegel with Herbart, Bradley with James. The systems' difficulty comes, as Wundt has aptly pointed out (see above § 10), from a single concept being attempted to be made into everything, although it has another concept for its correlate. Such precisely is the relation between unity and manifold. In the religious circle of representations, which in the last resort involuntarily follows the same laws as all other human thought-life, the relation of opposition spoken of leads to the formation of two different concluding concepts, each of which would presuppose the possibility of absolute totality: the concept of God and the concept of the world. The real concluding concept would here have to be a concept that could connect both these concepts — just as in Plato it would be a concept that could connect Ideas and phenomena, and in Spinoza a concept that could connect substance and modes.[6] If such a concept were to be capable of being formed, it would have to be a typical individual concept. For the existence in which we live stands as a *unicum*, a great individual whose characteristic peculiarity is determined by the content of the laws actually holding and by the constitution of the elements actually given. Spinoza is perhaps the one who has seen this most clearly. For he maintains that God (deus s. substantia s. natura) cannot come under any general concept (Ep. 50), and that the whole universe is one single great individual (Ethica II, Lemma 7, Schol.). It is therefore that in his doctrine of cognition he will not remain at the general laws, but demands a *scientia intuitiva* (a synthetic intuition), that is to say a cognition that is at once universal and individual. Every attempt at a scientific world-view aims at forming a typical individual concept containing a characterization of existence — this existence, which at once yields subject-matter for our thought and is itself a whole in which our thought is a part. Here

[6] Cf. my *Religionsfilosofi*^{2}, p. 59–62.

lies the root of the cosmological (metaphysical) problem. —

Besides the opposition between unity and manifold there makes itself felt, in the case of every whole, an opposition between value and reality, since the inner elements and the outer circumstances can both support and hinder its subsistence. The whole's subsistence can depend especially on the harmony and energy of the inner elements; but the germ of dissolution can often lie precisely in the constitution of these elements and their mutual relation. And it stands likewise with the outer circumstances. Value depends on the relation of the inner elements and the outer circumstances to the whole, when the latter's subsistence is regarded as the fundamental value. Since what is valuable is also a reality, it is really an opposition between a valuable reality and all other (indifferent or hindering) reality that we have here before us. The question here becomes whether we can form a concept that can connect both into a whole. Here lies the root of the religious problem.

Table of Categories

(Cf. Den menneskelige Tanke 1910)

I. Fundamental Categories.

1. Synthesis — Relation.
2. Continuity — Discontinuity.
3. Likeness — Difference.
    a. Chaotic difference-series.
    b. Indeterminately varying difference-series.
    c. Regularly varying difference-series.
    d. Identically varying difference-series.
        e. Progressive difference-series.
        f. Partial identity-series.
        g. Reciprocity-series.
        h. Absolute identity-series.
        (e–h:) Rational series.

II. Formal Categories.

1. Identity.
2. Analogy (reduction of relations of quality).
    a. Time. b. Number. c. Degree. d. Place.
3. Negation.
4. Rationality.

III. Real Categories.

1. Causality.
2. Totality.
3. Development.

IV. Ideal Categories (Value-concepts).

1. Formal relations of value.
2. Real relations of value.